

Stability Analysis and Optoelectronic Properties of Mg_3ZBr_3 ($\text{Z} = \text{As}, \text{Sb}, \text{Bi}$) Perovskites for Evaluating the Performance in PIN Photodiodes

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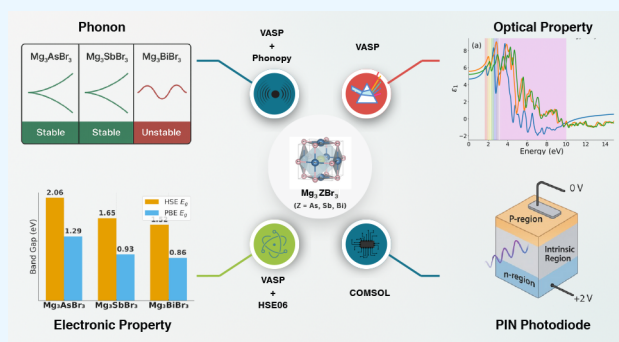
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ABSTRACT: The toxicity and stability issues of lead-based perovskites motivate nontoxic, durable alternatives. This work examines lead-free Mg_3ZBr_3 ($\text{Z} = \text{As}, \text{Sb}, \text{Bi}$) halide perovskites as optoelectronic materials, with an emphasis on Mg_3AsBr_3 and Mg_3SbBr_3 . Mg_3AsBr_3 and Mg_3SbBr_3 perovskites are predicted from first-principles calculations to crystallize in the cubic $Pm\bar{3}m$ phase, and indirect gaps of 2.0645 eV for Mg_3AsBr_3 , 1.6533 eV for Mg_3SbBr_3 , and 1.5226 eV for Mg_3BiBr_3 were observed with a hybrid functional. Optical spectra show a rise in absorption above the gap and an increasing static dielectric response along $\text{As} \rightarrow \text{Sb} \rightarrow \text{Bi}$. Phonon dispersion lacks imaginary branches for Mg_3AsBr_3 and Mg_3SbBr_3 , indicating the dynamical stability of these two materials under consideration, and exhibits large mode anharmonicity (Grüneisen signatures), consistent with soft-lattice heat transport trends. It has been found that moving down the pnictogen series expands the lattice and lowers the Goldschmidt tolerance factor, which, together with enhanced pnictogen–Br p-orbital hybridization and stereochemically active ns^2 lone pairs (Sb, Bi), narrows the bandgap and elevates the optical dielectric response. Elastic analyses confirm Born stability and moderate stiffness, with Hill-averaged bulk moduli decreasing from ~ 44 GPa (Mg_3AsBr_3) to ~ 35 GPa (Mg_3BiBr_3). Drift–diffusion p–i–n simulations qualitatively track band-edge-limited spectra, aligning with the computed gaps. Together, these results position Mg_3AsBr_3 and Mg_3SbBr_3 as lead-free candidates for stable thin-film photodiode and photovoltaic applications.



INTRODUCTION

Metal–halide perovskites have proven to be one of the most transformative material families in modern optoelectronics and are revolutionizing photovoltaic (PV) technologies. The power conversion efficiencies (PCEs) of perovskite solar cells (PSCs) have reached over 27% in 2025.¹ Previously, the PCE was first reported to be 3.8% in 2009, competing with crystalline silicon while offering advantages in solution processability, tunable bandgaps, and low-temperature fabrication.^{2,3} Though perovskites have been studied a lot both theoretically and experimentally because of high absorption coefficients ($>10^4 \text{ cm}^{-1}$), ambipolar charge transport, long carrier diffusion lengths ($>1 \mu\text{m}$), and high defect tolerance, it has been a point of major concern to find functional perovskites which are nontoxic to the environment.^{4–6} Additionally, instability under moisture, heat, and light limits operational lifetimes, necessitating encapsulation strategies that increase costs.^{7–9} To address these issues, lead-free halide perovskites have drawn attention because of their stability, excellent performance in devices, and eco-friendly attributes. Investigations into these materials are underway for a variety of applications that extend beyond solar cells, such as energy harvesting, radiation

detection, light-emitting diodes (LEDs), photodetectors, displays, photocatalysis, resistive memory devices, and sensors.^{10–12} In parallel, a growing body of first-principles and device-simulation studies on inorganic A_3BX_3 -type halide perovskites with group 15 cations (e.g., As, Sb, Bi) and alkaline-earth metals has begun to reveal favorable combinations of stability, tunable bandgaps, and high photovoltaic performance.^{13–17}

One of the emerging classes of lead-free perovskites is the A_3ZX_3 family, where A and Z represent large and small cations (Mg^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} , Sb^{3+} , As^{3+} , P^{3+}), respectively, and X represents F^- , Cl^- , Br^- , or I^- , which allow compositional flexibility to tailor electronic and structural properties.^{16,18} In an extensive study, Feng and Zhang reported the bandgaps of the A_3MX_3 ($\text{A} = \text{Mg}, \text{Ca}, \text{Sr}, \text{Ba}$; $\text{M} = \text{N}, \text{P}, \text{As}, \text{Sb}$; $\text{X} = \text{F}, \text{Cl}$,

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Br, I) materials for deciding the efficient materials in photovoltaics based on the theoretical efficiency and thermal stability.¹⁹ More recently, detailed first-principles calculations on A_3BX_3 ($A = \text{Ca, Sr}$; $B = \text{P, As}$; $X = \text{I, Br}$) have shown that these inorganic halide perovskites are direct-bandgap semiconductors with bandgaps at the Γ point in the range 1.26–2.63 eV and exhibit optical absorption of $(2\text{--}5) \times 10^4$ a.u. in the optical region below 3.2 eV.¹⁶ Complementary first-principles calculations on Ba_3MX_3 ($M = \text{As, Sb}$; $X = \text{Cl, Br, I}$) have demonstrated that all compounds are structurally stable, that six of them possess direct bandgaps in the range 1.35–1.65 eV, and that the fundamental gap can be tuned by X-anion substitution. Low carrier effective masses, high solar absorption, weak reflectance and energy loss, and ultrahigh theoretical single-junction conversion efficiencies of 31.9% for Ba_3MI_3 ($M = \text{As, Sb}$) and 29–30% for the remaining four compounds were reported, leading to the identification of Ba_3AsI_3 and Ba_3SbI_3 as particularly promising absorber layers.¹⁶ These advances underscore the potential of $A_3\text{ZX}_3$ -type perovskites with group 15 cations as efficient, lead-free photovoltaic materials.

Bismuth (Bi^{3+})-based double perovskites (e.g., $\text{Cs}_2\text{AgBiBr}_6$) offer better stability but suffer from indirect bandgaps and low carrier mobility, limiting PCEs to <5% as reported by Slavney et al.²⁰ Zhang and coauthors showed that stability can be improved with defect engineering and dimensional reduction (2D/3D heterostructures), yet efficiencies remain quite low compared to lead-based counterparts.²¹ In parallel, first-principles calculations on A_3BiI_3 ($A = \text{Cd, Mg, Ca, Sr, Ba}$) and Ba_3SbI_3 have revealed that all A_3BiI_3 compounds crystallize in a face-centered cubic lattice, with thermodynamic stability increasing from Cd_3BiI_3 to Ba_3BiI_3 according to their formation energies, and that both elastic constants and phonon dispersions confirm their mechanical and dynamic stability.¹³ The same study showed that Cd_3BiI_3 is metallic, Mg_3BiI_3 is a narrow-bandgap semiconductor with $E_g \approx 0.409$ eV, and Ca_3BiI_3 , Sr_3BiI_3 , and Ba_3BiI_3 are direct-bandgap semiconductors; in particular, Ba_3BiI_3 exhibits a direct bandgap of 1.309 eV and strong optical absorption ($\alpha > 10^5 \text{ cm}^{-1}$) in the 400–700 nm range, while Br substitution in $\text{Ba}_3\text{BiI}_{3-x}\text{Br}_x$ ($x = 0\text{--}3$) yields thermodynamically stable compounds with tunable bandgaps of 1.3–1.6 eV suitable as light-absorbing layers.¹³ On the other hand, tin (Sn^{2+})-halide perovskites, such as CsSnI_3 , initially showed promise with bandgaps near 1.3 eV and PCEs up to 11.57%.²² However, Sn^{2+} oxidizes rapidly to Sn^{4+} , creating vacancies that degrade performance.²³ The bandgap of perovskites is dependent on the BX_6 octahedral framework;²⁴ it has been shown that in MAPbI_3 , the Pb–I–Pb bond angles ($165^\circ\text{--}180^\circ$) and octahedral tilting dictate carrier mobility and bandgap.^{25,26} From the work by Zhang and Feng, it can be observed that substituting Pb^{2+} with smaller Mg^{2+} induces lattice contraction, altering the tilting dynamics.²⁷ Experimental X-ray diffraction confirms this lattice contraction—reflected by a $0.05\text{--}0.07^\circ$ shift of the (110) and (220) reflections—while preserving the perovskite phase, thereby validating the theoretical expectations.²⁸ The sensitivity of related antiperovskite derivatives X_3AsCl_3 ($X = \text{Mg, Ca, Sr, Ba}$) to external pressure further highlights the importance of structural tuning: first-principles calculations under hydrostatic pressures from 0 to 4 GPa reveal linear changes in structural and optoelectronic properties, with Ba_3AsCl_3 showing a decrease of 0.37 eV in bandgap and 0.251 Å in lattice constant at 4 GPa and achieving a theoretical conversion efficiency

exceeding 30% under moderate pressure, indicating its promise for optoelectronic devices under compressed epitaxial strain.¹⁵

Furthermore, a PCE of 24.73% has been observed for Ca_3NCl_3 as the absorber, which demonstrated a good response to visible spectrum absorption.²⁹ A recent first-principles investigation of the same lead-free Ca_3NCl_3 perovskite under pressure variations from +6% to –6% showed that it is thermodynamically and dynamically stable over this range, with charge-density maps indicating both ionic and covalent bonding features; the unstressed compound possesses a direct bandgap of 1.66 eV (PBE), 2.32 eV (HSE), and 1.48 eV (with spin–orbit coupling) at the Γ point, while compressive strain (–6%) reduces the bandgap to 1.37 eV (PBE), 1.93 eV (HSE), and 1.08 eV (SOC) and tensile strain (+6%) slightly increases it to 1.92 eV (PBE), 2.60 eV (HSE), and 1.73 eV (SOC), leading to corresponding red shifts and blue shifts in the absorption peaks.¹⁷ The same work reported enhanced optical absorption, loss function, conductivity, dielectric function, refractive index, and reflectivity under applied pressure as well as increased elastic constants, elastic moduli, ductility, machinability index, bulk and shear moduli, Pugh's ratio, Young's modulus, and anisotropy, making Ca_3NCl_3 more appropriate for use in solar absorbers, surgical tools, and optoelectronic devices.¹⁷ In another study, Apurba et al. demonstrated the direct bandgap and stability of Mg_3SbX_3 ($X = \text{I, Br, Cl, F}$) as a promising material for photovoltaic applications.³⁰ Magnesium (Mg^{2+})-based halide perovskites therefore represent a relatively underexplored but promising avenue. With an ionic radius comparable to Pb^{2+} , Mg^{2+} can occupy octahedral sites while offering nontoxicity and earth abundance.³¹ Early studies on CsMgX_3 ($X = \text{Cl, Br}$) revealed indirect bandgaps of 6.35 and 4.26 eV, respectively, which are suitable for UV photodetectors and dielectric applications.^{33,34} In addition to that, Sharma et al. predicted that CsMgBr_3 is an indirect bandgap material with moderate thermoelectric performance ($ZT > 0.79$).³⁴

This study addresses two critical gaps. By integration of As, Sb, and Bi into the material design, toxicity is avoided, aligning with global compliance standards. The second reason is that compositional tuning at the pnictogen site ($Z = \text{As, Sb, Bi}$) in Mg_3ZBr_3 enables control over the bandgap and dielectric response, allowing the cut-on wavelength and peak responsivity of Mg-based p–i–n photodiodes to be engineered without changing the device geometry. Motivated by recent demonstrations of structurally stable, direct-bandgap, and high-efficiency A_3BX_3 and Ba_3MX_3 perovskites containing group 15 cations as well as by the promising strain-tunable and device-level performance of Ca_3NCl_3 absorbers,^{13–17} we focus here on magnesium-based A_3ZX_3 halides. Using density functional theory (DFT), we systematically analyze Mg_3ZBr_3 ($Z = \text{As, Sb, Bi}$) perovskites with the aim of elucidating their structural, electronic, optical, and mechanical properties. Furthermore, we studied dynamical stability and assessed their viability as lead-free perovskites via PIN junction photodiode device simulations. This study provides a unified analysis of Mg_3ZBr_3 perovskites using a screened hybrid functional, bridging structural and optoelectronic properties and making them potential candidates in optoelectronic applications.

■ COMPUTATIONAL METHODOLOGY

All first-principles calculations were performed using the Vienna Ab Initio Simulation Package (VASP) v6.1.0.³⁵ VASP

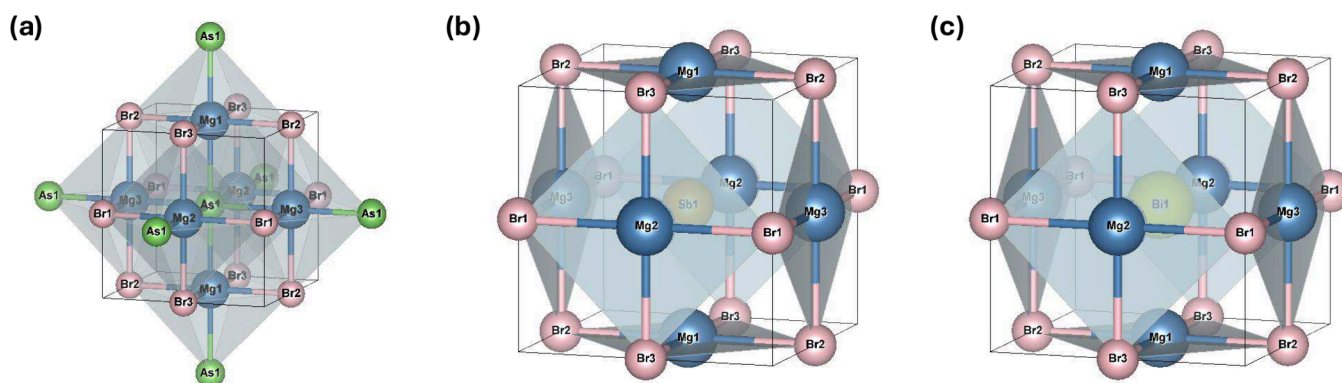


Figure 1. Crystal structures of (a) Mg_3AsBr_3 with As atoms (green) surrounded by Br atoms (pink) with Mg atoms (blue) at the corners, (b) Mg_3SbBr_3 with the Sb atom (yellow) at the center of the octahedron, and (c) Mg_3BiBr_3 with the Bi atom (yellow) at the center of the octahedron.

input files were generated and analyzed with the Python Materials Genomics (pymatgen) library³⁶ and the Atomic Simulation Environment (ASE).³⁷ We employed the projector augmented-wave (PAW) method³⁸ with VASP's standard PBE-PAW data sets. The valence electrons treated for each species were Mg $2p^63s^2$, As $4s^24p^3$, Sb $5s^25p^3$, Bi $6s^26p^3$, and Br $4p^5$. Exchange–correlation effects were described by the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation (GGA)³⁹ for geometry optimizations and most property calculations, while the screened hybrid functional of Heyd, Scuseria, and Ernzerhof (HSE06)⁴⁰ was used for high-accuracy structural relaxation, band structure, and optical property computations. For PBE calculations, a plane-wave kinetic energy cutoff of 400 eV was employed, whereas HSE06 calculations used a higher cutoff of 500 eV to ensure well-converged total energies and band structures; both values exceed the recommended minima for the employed PAW data sets. Brillouin zone integrations employed Monkhorst–Pack k -point meshes⁴¹ of $6 \times 6 \times 6$ for the primitive cubic Mg_3ZBr_3 unit cells, with commensurate mesh densities for larger supercells. The convergence of the total energy with respect to plane-wave cutoff and k -point sampling was explicitly verified for all three compositions, as shown in Figure S1. All structures were fully relaxed with respect to lattice parameters and atomic positions using the conjugate-gradient algorithm until residual Hellmann–Feynman forces fell below $0.01 \text{ eV } \text{\AA}^{-1}$ for each atom.

High-accuracy band structures were obtained by non-self-consistent evaluations using the HSE06 functional (25% exact exchange with a screening parameter of $0.2 \text{ \AA}^{-1}$).⁴⁰ Elastic properties were derived from stress–strain calculations at the PBE level by applying small deformations to the lattice and extracting the elastic stiffness tensor from the resulting stress responses. Optical properties, specifically the frequency-dependent dielectric function and absorption spectrum, were calculated within the independent-particle approximation using the PBE functional. Lattice-dynamical properties were evaluated using density functional perturbation theory (DFPT)⁴² as implemented in VASP. Phonon dispersion relations and phonon densities of states were then obtained by postprocessing the interatomic force constants with the Phonopy code.⁴³

Device simulations were performed in COMSOL Multiphysics (Semiconductor and EWFD Module) using a stationary drift–diffusion formulation under DC reverse bias. The device was treated as one-dimensional with a domain length

$h_{\text{dom}} = 1 \text{ }\mu\text{m}$. Ohmic p and n contacts were applied at the two ends. Gaussian dopant profiles were used at each contact with peak concentrations $N_A^{\text{max}} = N_D^{\text{max}} = 1 \times 10^{20} \text{ cm}^{-3}$ and junction depth $h_{\text{dom}} = 0.1 \text{ }\mu\text{m}$; the interior was intrinsic. Carrier transport and electrostatics obey the standard Poisson and continuity equations (eqs 10–13). Bulk recombination includes Shockley–Read–Hall (SRH), band-to-band radiative, and Auger channels. The SRH term employs a single midgap trap ($E_t = E_g/2$) with equal electron and hole lifetimes ($\tau_n = \tau_p = 1 \text{ ns}$). Radiative recombination is described by a halide-perovskite-typical coefficient $B = 1.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ and Auger recombination by coefficients $C_n = C_p = 1.0 \times 10^{-28} \text{ cm}^6 \text{ s}^{-1}$. No additional surface-recombination terms were introduced at the contacts (i.e., no explicit S_n, S_p), so recombination is dominated by these bulk processes. The structure is treated as a homojunction: the p, i, and n regions are all Mg_3ZBr_3 , and we therefore impose no band offsets at the internal p–i and i–n interfaces. The metal electrodes are modeled as ideal Ohmic contacts, with the carrier quasi-Fermi levels fixed by the applied terminal voltage.

RESULTS AND DISCUSSION

1. Structural Properties

The halide perovskites Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 crystallize in the simple-cubic space group $Pm\bar{3}m$ (No. 221), as shown in Figure 1. Within this topology, the pnictogen center occupies the body-center site $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, magnesium resides at the three face-centered positions $(\frac{1}{2}, \frac{1}{2}, 0)$, $(0, \frac{1}{2}, \frac{1}{2})$, and $(\frac{1}{2}, 0, \frac{1}{2})$, while bromine terminates the cube at $(0, 0, \frac{1}{2})$, $(\frac{1}{2}, 0, 0)$, and $(0, \frac{1}{2}, 0)$. The optimized lattice parameters and recent theoretical results are summarized in Table 1. As can be seen, our calculated lattice parameters are in very good agreement with the corresponding values reported in the literature. It is evident from Table 1 that both the lattice parameter and the unit-cell volume increase progressively from Mg_3AsBr_3 to Mg_3SbBr_3 and then to Mg_3BiBr_3 . This monotonic expansion reflects the progressive increase in the pnictogen ionic radius down group 15 of the periodic table.³¹ Complementary equation-of-state calculations, summarized in Figure 2, show equilibrium volumes V_0 of $24.18 \text{ \AA}^3 \text{ atom}^{-1}$ for Mg_3AsBr_3 (Figure 2a), $26.68 \text{ \AA}^3 \text{ atom}^{-1}$ for Mg_3SbBr_3 (Figure 2b), and $27.53 \text{ \AA}^3 \text{ atom}^{-1}$ for Mg_3BiBr_3 (Figure 2c). Concurrently, the binding energy per atom at equilibrium

Table 1. Optimized Cell Parameters (*a*) and Volumes (*V*) for Mg₃ZBr₃ (Z = As, Sb, Bi) Obtained with the PBE and Screened-Hybrid HSE06 Functionals^a

compound	<i>a</i> (Å)	<i>V</i> (Å ³)	method	<i>t</i>
Mg ₃ AsBr ₃	5.529	169.000	PBE	0.78
	5.480	164.566	HSE06	
	5.260	145.500 (93)	PBE	
Mg ₃ SbBr ₃	5.715	186.679	PBE	0.74
	5.660	181.321	HSE06	
	5.720	187.14 (30)	PBE	
Mg ₃ BiBr ₃	5.776	192.732	PBE	0.63
	5.718	186.953	HSE06	
	5.767	191.83 (16)	PBE	

^aThe tolerance factor *t* is listed for each Z. For each material, the first two rows list values calculated in this work, while the third row reports a literature bandgap for comparison.

becomes less negative: Mg₃AsBr₃ is the most strongly bound of the three, while Mg₃BiBr₃ exhibits the weakest binding. These observations indicate that substituting larger pnictogen atoms results in softer lattices and potentially lower bulk moduli, as further confirmed in Table 2. There is no phase transition observed within the pressure range from −4 to 4 GPa, as indicated in Figure 2. The lowest-energy structure throughout this range is the one at 0 GPa, which is consistent with previous studies of antiperovskites.^{15,32}

Although the unit-cell edge length grows steadily, the incremental volume increase between Sb and Bi members (186.7 Å³ → 192.7 Å³) is smaller than naively expected from ionic-radius arguments alone. Subtle changes in metal–halogen covalency and electronic configuration appear to temper the volumetric response, an observation echoed in earlier perovskite surveys.^{44,73} Electronic-structure insights follow logically from these metrics. A wider gap is anticipated for Mg₃AsBr₃, driven by the compact As⁵⁺ 4p manifold, whereas the Birch phase should exhibit improved carrier transport via spatially extended 6p orbitals.⁴⁴ Lone-pair activity on Sb³⁺ and Bi³⁺ may also induce second-order Jahn–Teller displacements that subtly tailor the band topology.⁴⁵ Such features, coupled with the benign defect chemistry familiar in halide perovskites,⁴⁶ argue for continued exploration of these Mg-based materials as environmentally friendly absorbers in photovoltaic and optoelectronic devices.⁴

Goldschmidt tolerance factors (*t*) provide a rapid gauge of perovskite formability. Employing Shannon's six-coordination radii ($r_{\text{Mg}^{2+}} = 0.72$ Å; $r_{\text{Br}^-} = 1.96$ Å; $r_{\text{As}^{5+}} = 0.46$ Å; $r_{\text{Sb}^{5+}} = 0.60$ Å; $r_{\text{Bi}^{3+}} = 1.03$ Å) gives *t* values of 0.78, 0.74, and 0.63 for Z = As, Sb, and Bi, respectively. All three lie below the canonical cubic interval (0.90–1.00),⁴⁷ forecasting octahedral tilts or lower-symmetry phases at ambient conditions.⁴⁸ The associated octahedral parameters $\mu = r_{\text{B}}/r_{\text{X}}$ are 0.23, 0.31, and 0.53, indicating that Bi substitution should deliver the most mechanically robust framework, whereas the As analogue approaches the threshold of geometric instability.⁴⁹ The tolerance-factor analysis signals latent structural flexibility that could be harnessed for phase-change sensing or chromic platforms.

2. Phonon and Lattice Dynamics

The phonon dispersion curves of cubic Mg₃ZBr₃ (Z = As, Sb, or Bi) are shown in Figure 3. It is clear from Figure 3a,c that Mg₃AsBr₃ and Mg₃SbBr₃ are dynamically stable in the cubic phase at 0 K, as evidenced by the absence of any imaginary

phonon modes in the calculated dispersions.⁵⁰ However, the presence of imaginary phonon bands for Mg₃BiBr₃ indicates that the structure is dynamically unstable and remains unstable under different volumes, as can be seen from Figure S5.

For Z = As, Sb, the phonon bandwidth (highest optical frequency) increases gradually from about 6.33 to 8.00 THz (Figure 3a,c). This trend is expected because heavier atoms and larger unit-cell volumes generally lead to lower vibrational frequencies.¹⁶ In fact, a clear decline in Debye temperature Θ_{D} (which scales with the acoustic phonon frequency) is observed as the atomic mass increases from As to Sb to Bi, which can be verified from the Hill-averaged mechanical properties in Table 2. The acoustic branches near the Brillouin zone center (Γ) become progressively less steep in going from As to Bi, indicating a decrease in sound velocity for the heavier compounds. This is consistent with the heavier average atomic mass and expanded lattice of Mg₃BiBr₃, which results in lower elastic wave speeds.⁵¹ The phonon bandwidth analysis of Figure 3b,d,f reveals a systematic reduction in the frequency span of the phonon branches in going from Mg₃AsBr₃ to Mg₃SbBr₃ to Mg₃BiBr₃. Specifically, Mg₃AsBr₃ exhibits the largest bandwidths (up to ~3.0 THz), while Mg₃SbBr₃ and Mg₃BiBr₃ show progressively narrower distributions, with the latter exhibiting the smallest overall dispersion. This trend reflects the increasing atomic mass of the pnictogen atom (As → Sb → Bi), which leads to a softening of the lattice vibrations and a corresponding reduction in phonon frequencies.

Each phonon dispersion in Figure 3a,c,e shows three acoustic modes (originating at Γ) and a manifold of optical modes at higher frequencies. There is a noticeable separation between the acoustic and optical phonon frequencies, especially pronounced for Mg₃SbBr₃ and Mg₃BiBr₃ compounds, where the highest acoustic mode ends below ~2 to 4 THz and a gap appears before the optical modes around 3–4 THz. This partial phonon bandgap arises because the heavy cation (Sb or Bi) and the Mg sublattice have much lower vibrational frequencies (acoustic and low optical modes), whereas the lighter atoms, especially Mg in all compounds, participate in the high-frequency optical modes, as observed in Figure 4a,c,e. The phonon density of states (DOS) in Figure 4a,c,e reflects this partitioning: the low-frequency region vibrations of the heavy Z atoms is observed, while the high-frequency peaks correspond largely to motions of the lighter Mg and Br atoms.⁵² For example, in Mg₃BiBr₃, the Bi atoms (the heaviest species) contribute primarily to the DOS below ~3 THz (including the flat, low-energy optical modes and acoustic modes). In contrast, the sharp DOS features above 4 THz are mainly due to Mg–Br stretching motions, with Bi remaining relatively immobile. A similar pattern is seen in Cs–Pb halide perovskites, where the heaviest cation (Cs) contributes to the lowest-frequency modes, and lighter anion vibrations dominate the high-frequency optical modes.⁵² This distribution of phonon modes has implications for thermal properties: a large fraction of low-frequency modes (as in the Bi compound) means more modes are thermally occupied at a given temperature, boosting the heat capacity and entropy at lower *T* values compared to a system with fewer low-frequency modes.

Figure 4b,d,f presents the Grüneisen parameters γ_i for each phonon mode, which measure the sensitivity of phonon frequency to volume changes. By definition, $\gamma_i = -\frac{\partial \ln \omega_i}{\partial \ln V}$ for mode *i*, so modes that soften rapidly upon expansion have

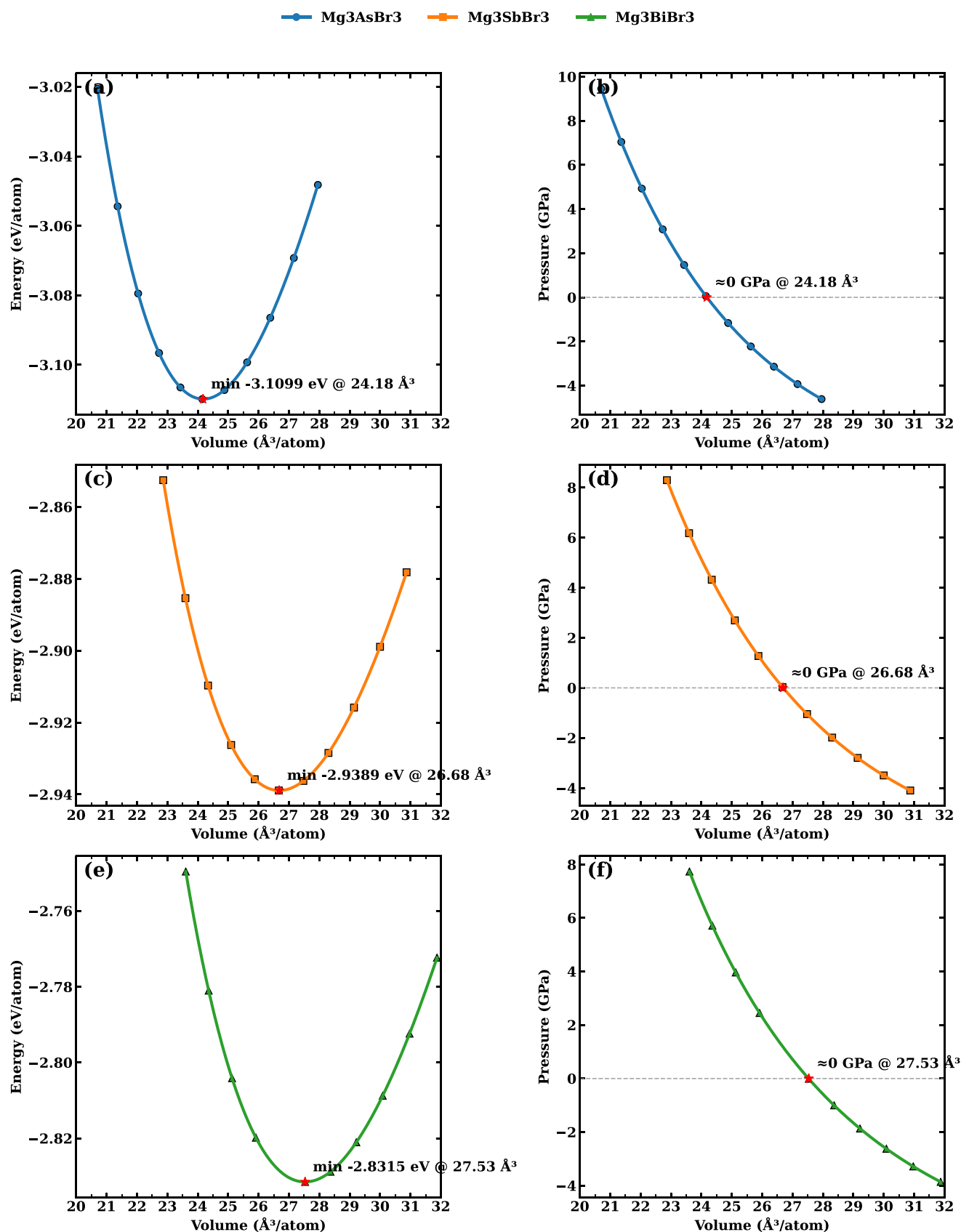


Figure 2. Equation-of-state (EOS) analysis for the Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 perovskites. (a, c, e) Total energy per atom versus volume, respectively. The red stars indicate the energy minima. (b, d, e) Corresponding pressure–volume curves. The red stars indicate the zero-pressure crossings that coincide with V_0 . The dashed gray lines denote zero pressure.

Table 2. Hill-Averaged Polycrystalline Mechanical Parameters and Principal Single-Crystal Elastic Constants of Mg_3ZBr_3 ($\text{Z} = \text{As, Sb, Bi}$)

compound	elastic constants (GPa)					polycrystalline moduli and derived parameters							
	C_{11}	C_{12}	C_{44}	$C_{11} - C_{12}$	$C_{11} + 2C_{12}$	B	E	G	H_V	ν	B/G	ζ	Θ_D (K)
Mg_3AsBr_3	84.485	23.447	28.033	61.038	131.379	43.793	71.273	29.002	3.323	0.229	1.510	0.919	314.8
Mg_3SbBr_3	74.694	20.586	22.460	54.108	115.866	38.622	60.050	24.197	2.814	0.241	1.596	0.830	276.6
Mg_3BiBr_3	68.824	18.717	20.448	50.105	106.258	35.419	55.275	22.181	2.598	0.240	1.597	0.816	242.9

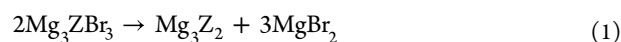
large positive γ . We computed γ_i for each mode via the quasi-harmonic approximation, using finite differences of phonon frequencies from volumes adjusted by $\pm 1\%$, $\pm 1.5\%$, and $\pm 2\%$ around equilibrium. The mode Grüneisen spectra (Figure 4b,d,f) reveal notably large Grüneisen values for the low-frequency branches, and the frequency spectrum average value $\bar{\gamma}$ is about 2.7 for Mg_3AsBr_3 , 3.2 for Mg_3SbBr_3 , and 2.8 for Mg_3BiBr_3 (dashed lines in Figure 4b,d,f). The heat-capacity-weighted averages are similarly high (e.g., $\gamma_{C_V} \approx 2.7\text{--}3.3$ at 300 K; dotted lines in Figure 4b,d,f). These γ values greatly exceed those of stiff, weakly anharmonic crystals (for instance, most rock-salt halides have $\gamma \approx 1.5\text{--}1.9$), indicating that Mg_3ZBr_3 perovskites are strongly anharmonic. Physically, a large Grüneisen parameter signals that phonon frequencies are very sensitive to lattice expansion, which is a hallmark of soft bonding and pronounced anharmonicity (e.g., bonding involving stereochemically active lone pairs, as in Bi^{3+} , tends to produce giant anharmonic phonon softening⁵²). The present compounds indeed have stereochemically active ns^2 cations (Sb^{3+} , Bi^{3+} at the B-site position, formally analogous to $\text{Sb}^{3+}/\text{Bi}^{3+}$ with lone pairs), which likely contribute to the large γ values via anharmonic rattling or off-centering tendencies.

Strong lattice anharmonicity and large Grüneisen parameters generally correlate with low lattice thermal conductivity since phonon–phonon Umklapp scattering rates increase with γ and phonon velocities decrease in soft lattices. According to Slack's theory of lattice thermal conductivity, materials with high Debye temperature, low average γ , and simple crystal structures will exhibit high thermal conductivity. The Mg_3ZBr_3 perovskites violate these criteria—they have relatively low Debye temperatures (especially for $\text{Z} = \text{Bi}$), large Grüneisen constants $\gamma \sim 3$, and a complex seven-atom primitive cell ($n = 7$)—and thus can be expected to possess quite low lattice thermal conductivities. Qualitatively, a high γ reduces the phonon mean free path via strong anharmonic scattering, and the small sound velocities in these soft, heavy-mass lattices further limit heat transport.⁵¹ This is consistent with the behavior of other halide perovskites: for example, CsPbBr_3 single crystals have $\kappa_L \approx 0.4\text{--}0.5 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature, and $\text{Cs}_2\text{AgBiBr}_6$ was recently reported to have an ultralow κ_L of $\sim 0.2 \text{ W m}^{-1} \text{ K}^{-1}$.⁵³ We therefore expect that Mg_3SbBr_3 and Mg_3BiBr_3 will exhibit extremely low thermal conductivities, making them attractive for thermal barrier coatings or thermoelectric applications where heat flow suppression is desired.¹⁶

Finally, the thermodynamic properties obtained from the phonon calculations are summarized in Figure S2. The temperature dependence of the Helmholtz free energy $F(T)$, entropy $S(T)$, and constant-volume molar heat capacity $C_V(T)$ is shown for $\text{Z} = \text{As, Sb, and Bi}$ in Figure S2a–c, respectively. All three compounds obey the expected Debye–Einstein trends. At low temperatures ($T \ll \Theta_D$), C_V rises as $\sim T^3$ and S as $\sim T^3$ (as indicated by the initial slopes in Figure S2). Free

energy curves $F(T)$ in Figure S2a show Mg_3SbBr_3 having the lowest vibrational free energy (most negative) at low T , though $F(T)$ converges among the compounds at higher temperature as $-TS$ dominates. By 1000 K, the entropy of Mg_3BiBr_3 reaches on the order of $500 \text{ J mol}^{-1} \text{ K}^{-1}$, while that of Mg_3SbBr_3 is around $480 \text{ J mol}^{-1} \text{ K}^{-1}$ (Figure S2b). Around room temperature, C_V approaches a substantial fraction of the Dulong–Petit limit. Notably, Mg_3AsBr_3 and Mg_3BiBr_3 have a larger heat capacity at 300 K than Mg_3SbBr_3 , as evident from Figure S2c. At $T \gtrsim 300 \text{ K}$, the heat capacity $C_V(T)$ of each compound begins to saturate near the Dulong–Petit value, and correspondingly, $S(T)$ grows more slowly. In summary, the lattice dynamics of Mg_3ZBr_3 are characterized by (i) dynamic stability of the cubic perovskite structure (no soft-mode instabilities at 0 K for $\text{Z} = \text{As, Sb}$, whereas instability for $\text{Z} = \text{Bi}$ may exist, correlating with Bi's low Goldschmidt tolerance factor), (ii) a systematic softening of the phonon spectrum in going from As to Bi due to increasing atomic mass and lattice size, (iii) phonon modes divided into low-frequency (acoustic and low optical) modes mainly involving heavy cation motion and high-frequency modes involving lighter atoms, and (iv) unusually large Grüneisen parameters indicating strong anharmonicity. These findings highlight the important role of the Goldschmidt tolerance factor and atomic masses in governing the vibrational stability of halide perovskites as well as the interplay between phonon dispersion and anharmonicity in determining thermal transport. Future work should explore finite-temperature lattice dynamics (e.g., quartic anharmonic stabilization of any soft modes and high-temperature phase transitions) and possibly measure the thermal conductivity of these materials to validate the ultralow values predicted (on par with the most thermally resistive perovskites like $\text{Cs}_2\text{AgBiBr}_6$ ⁵³).

In addition to dynamical stability, we assessed the thermodynamic stability of Mg_3ZBr_3 ($\text{Z} = \text{As, Sb, Bi}$) against decomposition into MgBr_2 and Mg_3Z_2 . Using PBE formation energies from the Materials Project for MgBr_2 and Mg_3Z_2 together with our PBE energies for Mg_3ZBr_3 , we evaluated the reaction



In all cases this decomposition is strongly exothermic (approximately 4.6–5.1 eV per Mg_3ZBr_3 formula unit), placing Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 about 0.67, 0.66, and 0.73 eV/atom above the convex hull, respectively (Table 3). Thus, while Mg_3AsBr_3 and Mg_3SbBr_3 are dynamically stable, all three compositions are thermodynamically metastable relative to separation into MgBr_2 and Mg_3Z_2 .

3. Electronic Properties and Band Structure Analysis

DFT calculations yield indirect gaps along $\Gamma \rightarrow \text{R}$ for all three compounds with a monotonic decrease across the pnictogen series (Table 4): E_g^{HSE} falls from 2.0645 eV (Mg_3AsBr_3) to 1.6533 eV (Mg_3SbBr_3) to 1.5226 eV (Mg_3BiBr_3) which is in

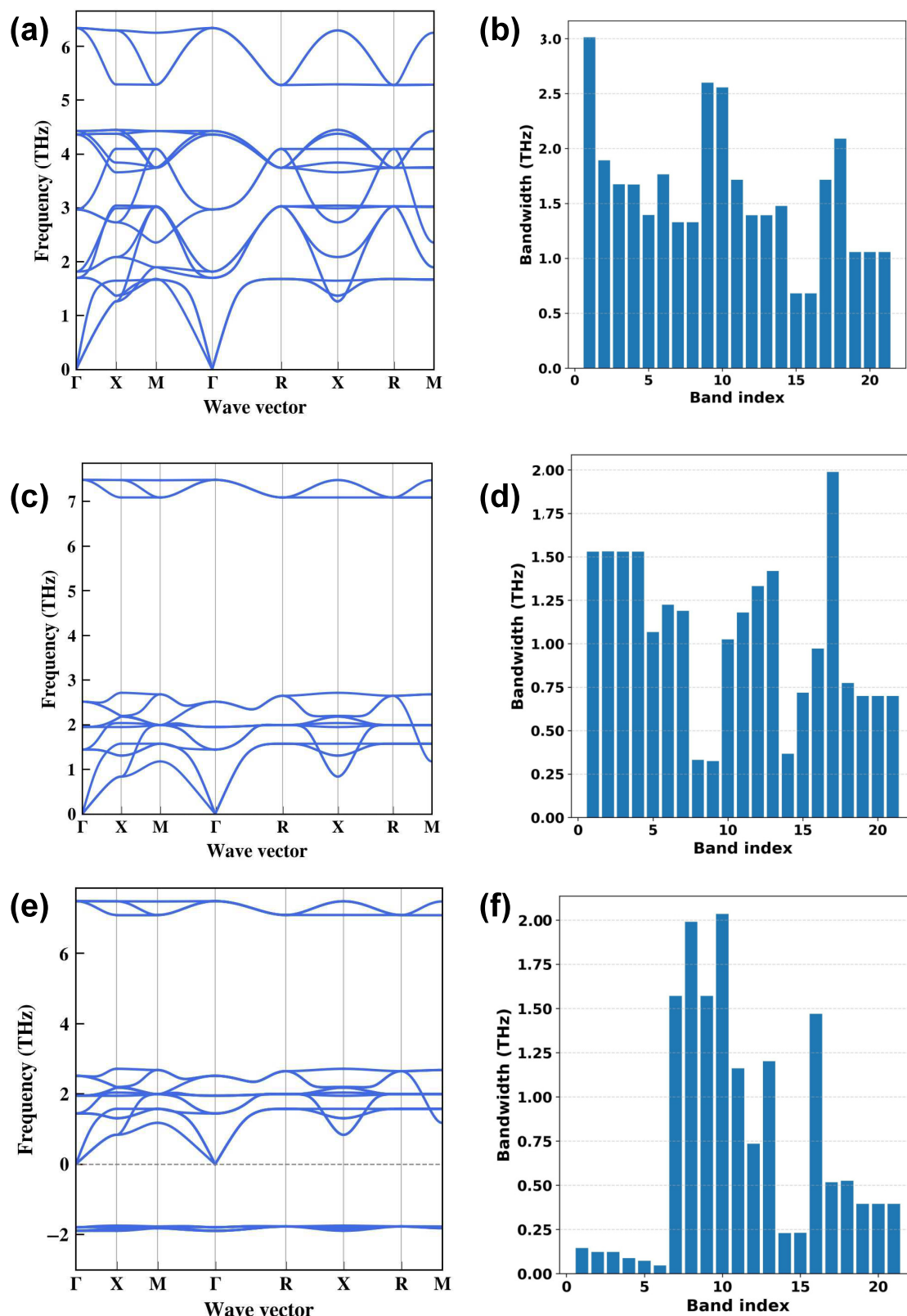


Figure 3. Phonon dispersion relations of cubic Mg_3ZBr_3 ($\text{Z} = \text{As}, \text{Sb}, \text{Bi}$) along high-symmetry directions and the frequency range that each branch spans along the computed path for (a, b) Mg_3AsBr_3 , (c, d) Mg_3SbBr_3 , and (e, f) Mg_3BiBr_3 .

good agreement with the previous theoretical studies as reported in Table 4. This decrease of bandgap aligns with the argument of Kim and coauthors for $\text{InP}/\text{InAs}/\text{InSb}$ using hybrid functionals.⁵⁴ Modulation of lone-pair stereochemical

activity in Bi compounds provides further support for the decrease of bandgap.⁵⁵ A similar down-group-V sequence was reported for NaZnX ($\text{X} = \text{P}, \text{As}, \text{Sb}, \text{Bi}$) associated with nontrivial band topology.⁵⁶ The bandgap range (1.5–2.1 eV)

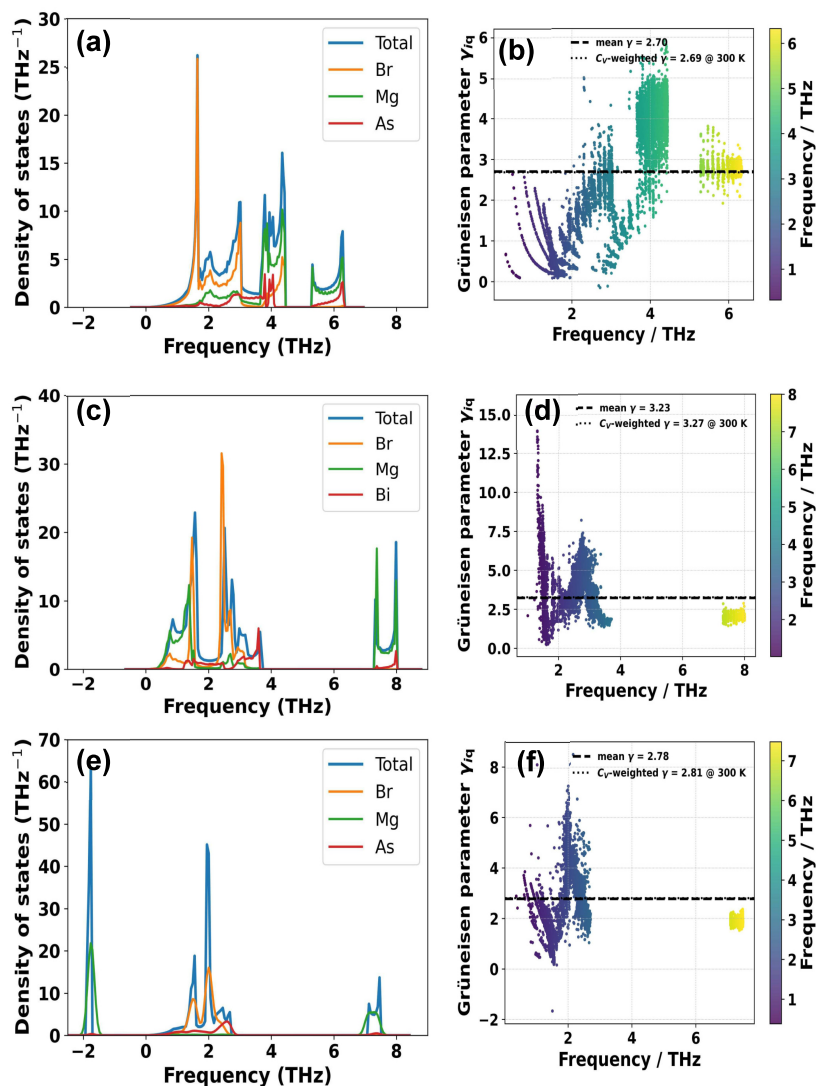


Figure 4. (a, c, e) Phonon DOS and PDOS of Mg₃ZBr₃ with (a) Z = As, (c) Z = Sb, and (e) Z = Bi and (b, d, f) mode Grüneisen parameter γ_{iq} vs mode frequency for (b) Z = As, (d) X = Sb, and (f) Z = Bi. In the Grüneisen plots, each point corresponds to a phonon mode (color-coded by frequency for clarity). The average Grüneisen parameter (dashed line) and the heat-capacity-weighted average at 300 K (dotted line) are indicated.

Table 3. Decomposition Energetics of Mg₃ZBr₃ (Z = As, Sb, Bi) with Respect to MgBr₂ and Mg₃Z₂ at 0 K (PBE)^a

compound	dominant decomposition reaction (0 K)	ΔE_{rxn} (eV/f.u.)	ΔE_{rxn} (eV/atom)	$E_{\text{above hull}}$ (eV/atom)
Mg ₃ AsBr ₃	2Mg ₃ AsBr ₃ → Mg ₃ As ₂ + 3MgBr ₂	−4.683	−0.669	0.67
Mg ₃ SbBr ₃	2Mg ₃ SbBr ₃ → Mg ₃ Sb ₂ + 3MgBr ₂	−4.60	−0.657	0.66
Mg ₃ BiBr ₃	2Mg ₃ BiBr ₃ → Mg ₃ Bi ₂ + 3MgBr ₂	−5.13	−0.733	0.73

^a ΔE_{rxn} is per formula unit of Mg₃ZBr₃; $E_{\text{above hull}}$ is the corresponding energy above the MgBr₂ + Mg₃Z₂ convex hull.

Table 4. Electronic Bandgaps, Carrier Mobilities, and Direction-Resolved Effective Masses of Mg₃ZBr₃ (Z = As, Sb, Bi)^a

material	$E_{\text{g}}^{\text{HSE}}$ (eV)	$E_{\text{g}}^{\text{PBE}}$ (eV)	μ_{n} (cm ² V ^{−1} s ^{−1})	μ_{p} (cm ² V ^{−1} s ^{−1})	k path	m_{e}^*/m_0	m_{h}^*/m_0
Mg ₃ AsBr ₃	2.0645	1.2907	57.86	57.86	R → Γ	0.336	−1.096
					Γ → R	0.291	−0.959
Mg ₃ SbBr ₃	2.353	1.485 ⁹⁰	57.86	30.22	R → Γ	0.335	−0.989
	1.6533	0.9259			Γ → R	0.253	−0.648
	2.0178	0.9578 ³⁰					
	1.5226	0.8587					
Mg ₃ BiBr ₃	1.5226	0.8587	57.86	59.20	R → Γ	0.335	−0.950
					Γ → R	0.238	−0.565
	1.626	1.071 ¹⁶					

^aFor each material, the top two rows list values calculated in this work, while the bottom row reports a literature bandgap for comparison.

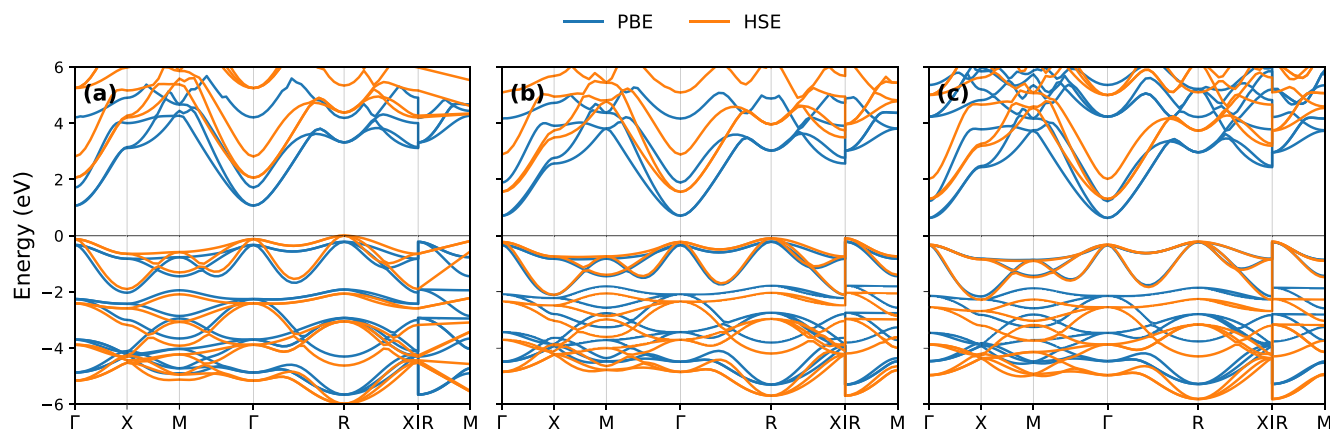


Figure 5. Electronic band structures of (a) Mg_3AsBr_3 , (b) Mg_3SbBr_3 , and (c) Mg_3BiBr_3 computed with PBE (blue dashed) and HSE06 (red solid). The horizontal dotted line marks the Fermi level at $E = 0$ eV. The high-symmetry path is Γ -X-M- Γ -R-X|R-M. The fundamental gaps are $E_g^{\text{HSE}} = 2.0645$ eV ($E_g^{\text{PBE}} = 1.2907$ eV) for (a), $E_g^{\text{HSE}} = 1.6533$ eV ($E_g^{\text{PBE}} = 0.9259$ eV) for (b), and $E_g^{\text{HSE}} = 1.5226$ eV ($E_g^{\text{PBE}} = 0.8587$ eV) for (c).

Table 5. Material and Recombination Parameters Adopted in the Drift–Diffusion Simulations for the Mg_3ZBr_3 Family

property	symbol	unit	Mg_3AsBr_3	Mg_3SbBr_3	Mg_3BiBr_3
relative permittivity	ϵ_r	—	4.57	5.11	5.7
band gap (HSE06)	E_g	eV	2.06	1.65	1.52
valence-band effective DOS	N_v	cm^{-3}	2.78×10^{19}	2.4×10^{19}	2.2×10^{19}
conduction-band effective DOS	N_c	cm^{-3}	4.72×10^{18}	4.7×10^{18}	4.7×10^{19}
electron mobility	μ_n	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	57.86	30.22	59.2
hole mobility	μ_p	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	4.87	3.54	2.21
SRH lifetime (electrons)	τ_n	ns	1	1	1
SRH lifetime (holes)	τ_p	ns	1	1	1
radiative recombination coefficient	B	$\text{cm}^3 \text{s}^{-1}$	1.0×10^{-10}	1.0×10^{-10}	1.0×10^{-10}
Auger recombination coefficient	$C_n = C_p$	$\text{cm}^6 \text{s}^{-1}$	1.0×10^{-28}	1.0×10^{-28}	1.0×10^{-28}
refractive index (real part)	n	—	2.34	2.14	2.28
refractive index (imaginary part)	k	—	0.00	0.00	0.00

suits single-junction PV; Mg_3BiBr_3 (1.52 eV) and Mg_3SbBr_3 (1.65 eV) fall within the Shockley–Queisser limit,⁵⁷ although the indirect gap favors thicker absorbers.

Effective masses were obtained from the local band curvature using fits near the band edges:

$$m^* = \hbar^2 \left(\frac{d^2 E}{dk^2} \right)^{-1} \quad (2)$$

The values are listed in Table 4. All of the compounds show flat VBM dispersion (large $|m_h^*|$) and a more dispersive CBM at Γ (smaller m_e^*) (Figure 5).⁵⁸ This asymmetry points to higher electron than hole mobility, consistent with relations between dispersion and transport.⁵⁹ The mobility was calculated using eq 3:

$$\mu = \frac{\sqrt{8\pi} e \hbar^4 C}{3(m^*)^{5/2} (k_B T)^{3/2} E_1^2} \quad (3)$$

where μ denotes the carrier mobility (in $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$), C represents the longitudinal elastic modulus (such as C_{11}), m^* denotes the carrier effective mass, and E_1 corresponds to the longitudinal deformation potential. The calculated values are listed in Table 5.

The projected DOS identifies the orbital contributions in the electronic DOS. As observed from Figure S3a–c, in all three compounds, the valence band is dominated by Br 4p states with increasing pnictogen p character near the VBM; Mg's contribution near the edges is small. The fundamental

gap is a transition from Br 4p-dominated valence states to conduction states with pnictogen s/p and Mg s orbitals, consistent with prior perovskite studies.^{25,60} The indirect gaps affect the optical and electronic response: indirect absorption is weaker (phonon assistance required), while nonradiative recombination can be reduced, extending lifetimes.

Figure 6 shows the (100) plane electron localization functions (ELFs) of Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 . All

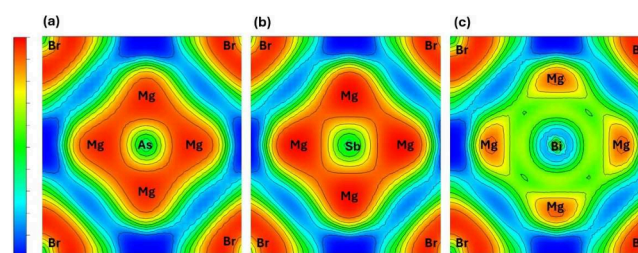


Figure 6. Two-dimensional ELF maps in the (100) planes of (a) Mg_3AsBr_3 , (b) Mg_3SbBr_3 , and (c) Mg_3BiBr_3 . Color scale: ELF = 0 (blue) to 1 (red).

three exhibit a pronounced ns^2 lone-pair basin at the pnictogen. The maximum ELF value decreases in going from As to Sb to Bi as observed from Figure 6a–c. Here outside the lone-pair basin, Mg–Br linkages have ELF < 0.2 (ionic), whereas Pn–Br bond lines retain intermediate ELF (~0.45–0.60), indicating mixed ionic–covalent character that strength-

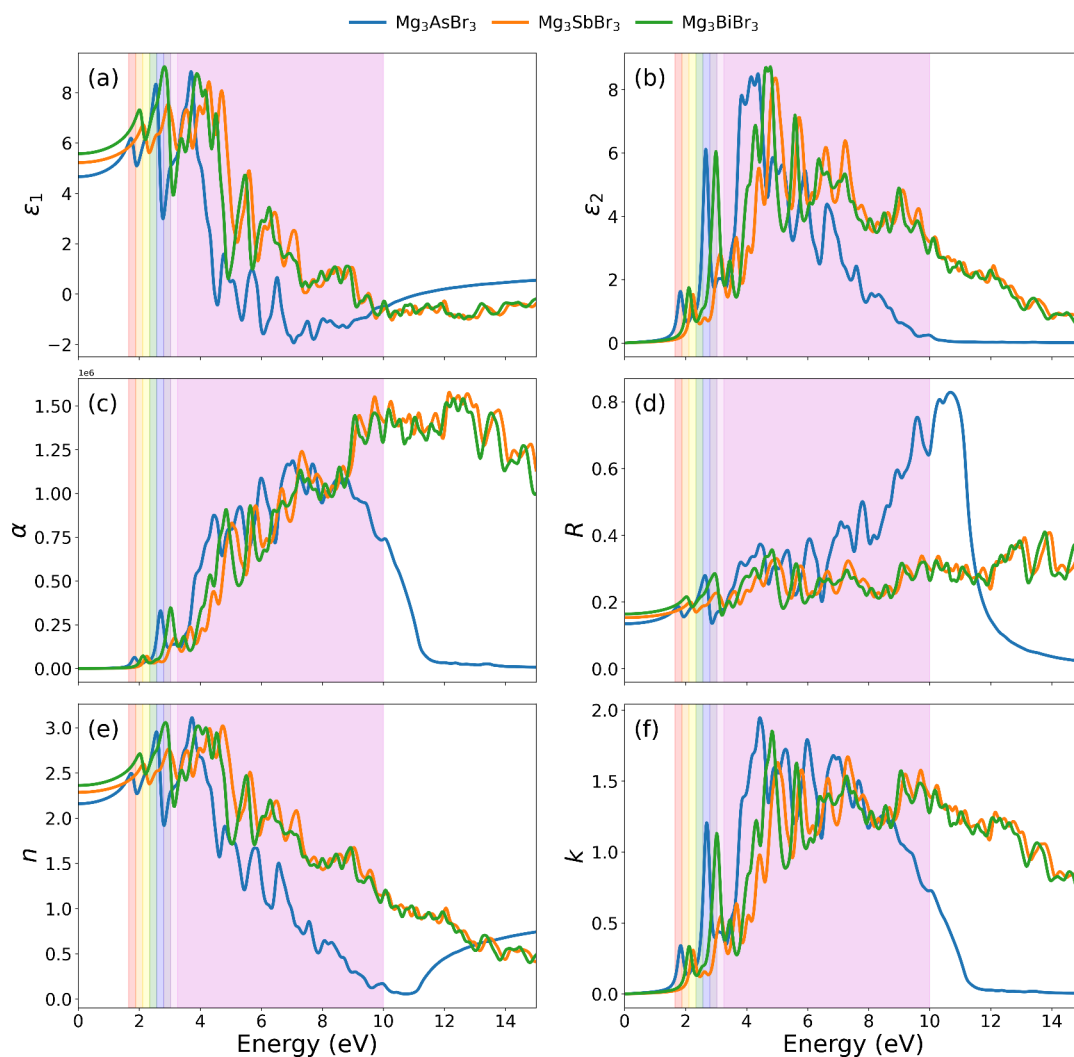


Figure 7. Calculated optical properties of Mg_3ZBr_3 ($\text{Z} = \text{As}, \text{Sb}, \text{Bi}$) perovskites as a function of photon energy: (a) real part of the dielectric function; (b) imaginary part of the dielectric function; (c) optical absorption coefficient; (d) optical reflectivity; (e) refractive index; (f) extinction coefficient. The colored vertical bands in the low-energy region represent the visible region, while the violet-shaded regions highlight the UV spectrum.

ens in going from As to Bi due to increased Pn p/s–Br p overlap.⁶¹ Bonding explains the bandgap trends as well. The Mg–Br network is largely ionic, while Pn–Br bonds show mixed character. In going from As to Bi, stronger Pn p/s–Br p hybridization increases antibonding weight at the CBM and narrows E_g . The stereochemically active ns^2 lone pairs on Sb and Bi add states near the band edges, a behavior well-documented in related Bi compounds.⁴⁵

To clarify the oxidation states of the pnictogen atoms in Mg_3ZBr_3 ($\text{Z} = \text{As}, \text{Sb}, \text{Bi}$), we performed Bader charge analysis of the DFT valence charge density. In all three compounds, the three Mg and three Br atoms are symmetry-equivalent, so we report their average Bader charges together with that of the pnictogen atom. The net Bader charge q_B on each atom was obtained as

$$q_B = Z_{\text{val}} - N_{\text{Bader}} \quad (4)$$

where Z_{val} is the number of valence electrons in the PAW potential (Mg, 2; Br, 7; As/Sb, 5; Bi, 15) and N_{Bader} is the integrated Bader electron population from the ACF.dat file. Positive q_B corresponds to cationic character (electron loss),

while negative q_B corresponds to anionic character (electron gain).

As summarized in Table S1, Mg carries a net positive charge of about $+1.76e$ to $+1.81e$, while Br carries approximately $-1.0e$, consistent with the expected Mg^{2+} and Br^- species. The pnictogen atoms exhibit significantly more negative charges with $q_B(\text{As}) \approx -2.38e$, $q_B(\text{Sb}) \approx -2.46e$, and $q_B(\text{Bi}) \approx -2.29e$. Although Bader charges are noninteger by nature, these values are close to the ideal -3 charge expected for pnictide anions and clearly distinguish the pnictogens from the halides. Combining the formal charge balance ($3 \times \text{Mg}^{2+} + 3 \times \text{Br}^-$) with the Bader analysis therefore supports the assignment of arsenic, antimony, and bismuth in Mg_3ZBr_3 as trivalent pnictide anions (Z^{3-} , i.e., oxidation state $-III$).

4. Optical Properties and Spectroscopic Analysis

The computed optical parameters include the complex dielectric function $\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega)$, absorption coefficient $\alpha(\omega)$, reflectivity $R(\omega)$, refractive index $n(\omega)$, and extinction coefficient $\kappa(\omega)$. These properties are fundamentally connected to the electronic structure and provide critical insights into the light–matter interactions essential for

applications in photovoltaics, photodetectors, and other optoelectronic devices.

Figure 7a,b displays the real (ϵ_1) and imaginary (ϵ_2) components of the dielectric function for all three compounds across the 0–15 eV energy range. The static dielectric constants ($\epsilon_1(0)$) for the three materials were determined to be approximately 4.8, 5.7, and 5.9 for Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 , respectively. This systematic increase with heavier pnictogen elements correlates with the decreasing bandgap, consistent with the Penn model.^{62,63}

Notably, in the real part (Figure 7a), all three materials show pronounced peaks in the 2–7 eV region, corresponding to critical points in their joint densities of states where significant interband transitions occur. Mg_3AsBr_3 displays sharper spectral features with more distinct oscillations compared with the other two compounds, which exhibit somewhat broadened spectral features. The dielectric function approaches asymptotic values at higher energies (>12 eV), indicating transitions to the free-electron-like regime. The imaginary part $\epsilon_2(\omega)$ (Figure 7b), which directly relates to the absorption of photons, exhibits significant absorption peaks after the 2 eV range. The onset of absorption is consistent with the calculated bandgaps: ~2.06 eV for Mg_3AsBr_3 , ~1.65 eV for Mg_3SbBr_3 , and ~1.52 eV for Mg_3BiBr_3 . Mg_3SbBr_3 and Mg_3BiBr_3 display somewhat similar spectral profiles. Based on the projected density of states analysis, the primary transitions can be attributed to Br 4p $\rightarrow$ pnictogen p/s transitions, with the exact energy dependent on the specific pnictogen element.⁶⁴

The absorption coefficient $\alpha(\omega)$ can be calculated from the complex dielectric function using the following equation:

$$\alpha(\omega) = \frac{2\omega}{c} \sqrt{\frac{\epsilon_1^2(\omega) + \epsilon_2^2(\omega) - \epsilon_1(\omega)}{2}} \quad (5)$$

This provides a quantitative measure of the light absorption capability of the materials. As shown in Figure 7c, all three compounds exhibit strong absorption in the visible and ultraviolet regions, with absorption coefficients reaching approximately $(0.8\text{--}1.5) \times 10^6 \text{ cm}^{-1}$ at higher photon energies. These characteristics align with the indirect bandgap nature of these materials, where phonon-assisted transitions are necessary for photon absorption near the band edge, resulting in relatively weak absorption near the onset. However, the absorption coefficients increase rapidly at energies well above the bandgap, where direct transitions become dominant.⁶⁴

The reflectivity $R(\omega)$ can be calculated from the complex dielectric function using the following equation:

$$R(\omega) = \left| \frac{\sqrt{\epsilon(\omega)} - 1}{\sqrt{\epsilon(\omega)} + 1} \right|^2 \quad (6)$$

This represents the fraction of incident light reflected at the surface of the material surface. The reflectivity spectra in Figure 7d show distinctive features that correlate with the dielectric function. In the low-energy region (near the bandgap), the reflectivity values are approximately 0.17, 0.185, and 0.19 for Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 , respectively, indicating that 17–19% of the incoming light is reflected from the surface. This relatively low reflectivity in the visible range is advantageous for maximizing light coupling into the absorber layer.⁴ The reflectivity increases significantly in regions of strong absorption, reaching maxima of approximately 0.82 for Mg_3AsBr_3 around 11 eV, while Mg_3SbBr_3 and

Mg_3BiBr_3 show lower maximum reflectivity values of approximately 0.35–0.40 in the high-energy region. A particularly interesting feature is the significantly higher reflectivity of Mg_3AsBr_3 compared to the other compounds in the 9–12 eV range, which suggests fundamentally different high-energy electronic transitions. This difference may be attributed to the distinct orbital character and energy positioning of the As 4p states compared to the heavier pnictogens' p states.⁶⁵

The refractive index $n(\omega)$ can be derived from the complex dielectric function using the following equation:

$$n(\omega) = \sqrt{\frac{\epsilon_1^2(\omega) + \epsilon_2^2(\omega) + \epsilon_1(\omega)}{2}} \quad (7)$$

This provides information about light propagation through the material, which is represented in Figure 7e. The calculated static refractive indices (at zero frequency) are approximately 2.2, 2.4, and 2.45 for Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 , respectively. This trend follows the Moss relation that $n^4 \cdot E_g$ is approximately constant across similar material families.⁶⁶ The moderate refractive indices (2.2–2.5) of the studied materials in the near-infrared region could be exploited for optical coating applications. The refractive index spectra show significant dispersion in the visible and ultraviolet regions, with maximum values of approximately 3.1, 3.0, and 3.0 for the As, Sb, and Bi compounds, respectively, occurring in the 3–5 eV range. At energies above 8 eV, the refractive index decreases substantially for all three materials, approaching values below 1 in regions of strong absorption, a phenomenon associated with anomalous dispersion near resonant absorption frequencies.⁶⁷

The extinction coefficient $\kappa(\omega)$, which represents the imaginary part of the complex dielectric function, is directly related to the absorption coefficient through the following expression:

$$\kappa(\omega) = \frac{\alpha(\omega)\lambda}{4\pi} \quad (8)$$

This provides a measure of the attenuation of light as it passes through the material. The extinction coefficient spectra closely mirror the trends observed in the imaginary part of the dielectric function. All three compounds in Figure 7f show maximum extinction coefficients in the 3–7 eV range with peak values of approximately 1.95, 1.65, and 1.60 for Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 , respectively. The extinction coefficient of Mg_3AsBr_3 drops to nearly zero above 12 eV, while the heavier pnictogen compounds maintain moderate extinction coefficients even at higher energies, suggesting different high-energy absorption mechanisms possibly related to deeper core-level transitions.⁶⁴

5. Elastic Properties and Mechanical Stability

The elastic properties of materials are fundamental characteristics that describe their mechanical responses to applied forces within the reversible deformation regime. These properties are essential for understanding structural stability, mechanical behavior, and potential applications in various technological domains. The investigated materials crystallize in the cubic perovskite structure with space group $Pm\bar{3}m$ (No. 221), exhibiting high symmetry that reduces the independent elastic constants to only three: C_{11} , which represents longitudinal deformation resistance; C_{12} , which describes the cross-coupling between perpendicular deformations; and C_{44} , which repre-

sents the resistance to shear deformation. Table 2 presents the calculated elastic constants for all three studied compounds. The elastic constants display a clear trend with all values decreasing in going from Mg_3AsBr_3 to Mg_3BiBr_3 , which can be attributed to the increasing atomic radius of the pnictogen element and the consequent expansion of the lattice parameters. This observation aligns with the well-established principles of bond length–strength correlation in crystalline solids, where longer bonds typically exhibit weaker bond strength, leading to reduced elastic constants. A similar trend has been observed in various perovskite systems, as reported by Ekuma et al.⁶⁸ and Savinov et al.⁶⁹

The mechanical stability of a crystalline material is governed by the Born stability criteria, which for cubic crystals require $C_{11} - C_{12} > 0$ (resistance to tetragonal shear deformation), $C_{11} + 2C_{12} > 0$ (resistance to uniform volume compression), and $C_{44} > 0$ (resistance to trigonal shear deformation). As evident from Table 2, Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 satisfy all of these criteria, confirming their mechanical stability. The positive eigenvalues of the stiffness matrix further corroborate this conclusion. The value of $C_{11} - C_{12}$ is 61.038 GPa for Mg_3AsBr_3 , 54.108 GPa for Mg_3SbBr_3 , and 50.105 GPa for Mg_3BiBr_3 , indicating substantial resistance to tetragonal deformations, which is important for maintaining structural integrity under anisotropic stress conditions. The ratio C_{12}/C_{11} , which represents the Cauchy pressure normalized by C_{11} , provides insight into the nature of interatomic bonding. The calculated values of the Cauchy pressure of 0.277 for Mg_3AsBr_3 , 0.276 for Mg_3SbBr_3 , and 0.272 for Mg_3BiBr_3 are relatively low, suggesting a significant degree of directional covalent character in their bonding. This observation is consistent with the negative Cauchy pressure ($C_{12} - C_{44}$) values of -4.6 GPa, -1.9 GPa, and -1.7 GPa for Mg_3AsBr_3 , Mg_3SbBr_3 and Mg_3BiBr_3 respectively, as mentioned in Table 2, further indicating predominant covalent bonding characteristics with some ionic contribution, as mentioned by Pettifor⁷⁰ and others in their seminal works on the relationship between Cauchy pressure and bonding nature. This is consistent with the results found in the ELF calculations shown in Figure 6. From the elastic constants, various important mechanical parameters can be derived, including bulk modulus (B), Young's modulus (E), shear modulus (G), and Poisson's ratio (ν). For polycrystalline materials, these properties are typically estimated using the Voigt, Reuss, and Hill approximations, which provide upper-bound, lower-bound, and average estimates, respectively. Table 2 summarizes the calculated Hill average values of these properties for the Mg_3ZBr_3 compounds.

The bulk modulus (B) represents the material's resistance to hydrostatic compression and is inversely proportional to the compressibility. The calculated bulk modulus values of 43.793 GPa for Mg_3AsBr_3 , 38.622 GPa for Mg_3SbBr_3 , and 35.419 GPa for Mg_3BiBr_3 are relatively moderate compared to those of typical oxide perovskites like BaTiO_3 (~ 162 GPa) and SrTiO_3 (~ 179 GPa) but higher than those of many halide perovskites such as CsPbI_3 (~ 19.8 GPa) as reported by Rakita et al.⁷¹ The observed decrease in bulk modulus in going from Mg_3AsBr_3 to Mg_3BiBr_3 can be attributed to the increased lattice parameter and, subsequently, weaker average bond strength. The bulk modulus can be related to the interatomic bonding strength through empirical relationships such as the one proposed by Cohen:⁷²

$$B = k \cdot d^{-3.5} \quad (9)$$

where d is the average bond length and k is a proportionality constant. The decreasing bulk modulus with an increasing pnictogen size follows this relationship, indicating the dominant role of bond length in determining compressibility in these materials.

The Young's modulus (E) represents the material's stiffness under uniaxial deformation, while the shear modulus (G) describes resistance to shape change under shear stress. Both moduli show a decreasing trend in going from Mg_3AsBr_3 to Mg_3BiBr_3 . This reduction is more pronounced than that observed for the bulk modulus, suggesting that the materials become particularly less resistant to directional (shear and tensile) deformations as the pnictogen size increases, a behavior that can be attributed to the greater polarizability of larger atoms leading to more easily distorted electron clouds under directional stress.⁷³ Poisson's ratio (ν), which represents the negative ratio of transverse to axial strain under uniaxial stress, provides valuable insights into the nature of interatomic forces. As can be seen from Table 2, calculated values are within the typical range for ionic and covalent solids (0.2–0.3). According to the classification scheme proposed by Greaves et al.,⁷⁴ materials with Poisson's ratios below 0.25 tend to exhibit more open structures with predominantly directional bonding, aligning with the covalent character suggested by the negative Cauchy pressure. The brittleness or ductility of materials can be assessed using Pugh's ratio (B/G), with a critical value of 1.75 typically distinguishing between brittle (< 1.75) and ductile (> 1.75) behavior. The calculated Pugh's ratios of 1.510 for Mg_3AsBr_3 , 1.596 for Mg_3SbBr_3 , and 1.597 for Mg_3BiBr_3 suggest that all materials exhibit brittle characteristics, although the higher values indicate a slight trend toward increased ductility with increasing pnictogen size. This trend can be rationalized by considering that the larger, more polarizable Sb and Bi atoms allow for greater structural flexibility and plastic deformation before bond breaking.⁷⁵ Though the B/G ratios of the studied materials are less than 1.75, they are very close to 1.75. There are studied materials for thin film applications,^{91,92} which have the B/G ratios similar to those of our studied materials. The brittle nature indicated by Pugh's ratios below 1.75 does necessitate careful handling during device fabrication to prevent fracture.⁴ The Cauchy pressure ($C_{12} - C_{44}$) provides an additional perspective on ductility, with negative values typically associated with directional bonding and brittle behavior, while positive values indicate more delocalized metallic bonding and ductility. The negative Cauchy pressures of these materials corroborate the brittle nature of these materials while suggesting a slight trend toward increased metallicity.⁷⁰ The Debye temperature (Θ_D), calculated from the average sound velocity using the quasi-harmonic Debye model, provides insights into the lattice thermal properties and phonon behavior. The estimated values that are mentioned in Table 2 are relatively low compared to those of typical oxide perovskites, indicating softer lattice phonons and potentially higher phonon scattering rates. This characteristic could be beneficial for thermoelectric applications where low thermal conductivity is desirable. The decrease in Debye temperature in going from Mg_3AsBr_3 to Mg_3BiBr_3 is consistent with the trend in elastic moduli and can be attributed to the lower-frequency vibrations associated with the heavier pnictogen atoms and weaker interatomic forces.⁷⁶ The trend of

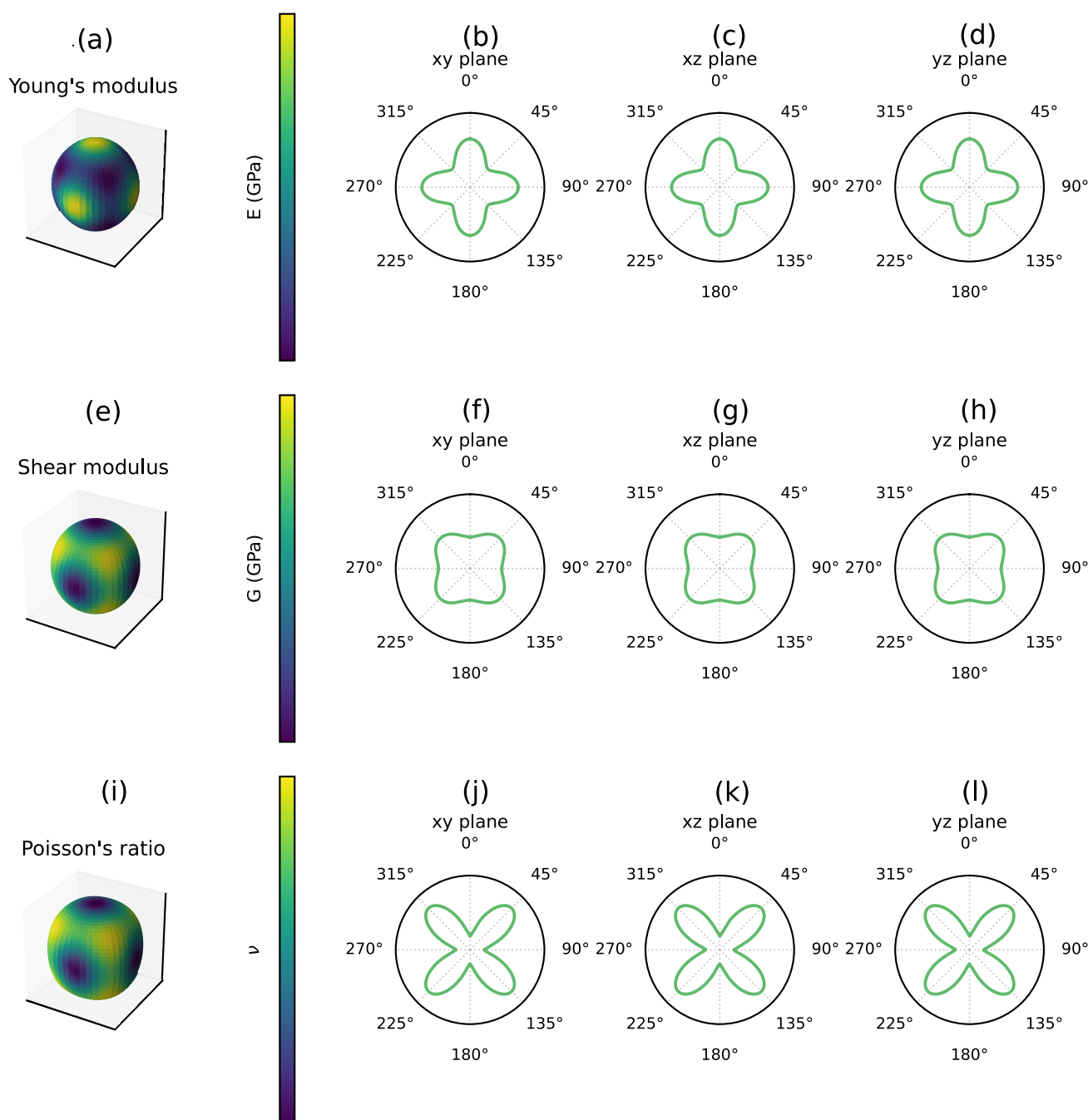


Figure 8. Orientation-dependent elastic response of cubic Mg_3AsBr_3 . (a) Three-dimensional surface and (b–d) two-dimensional polar projections of Young's modulus $E(n)$ in the XY , YZ , and YZ crystallographic planes, respectively. (e–h) Corresponding representations of the shear modulus $G(n)$. (i–l) Polar projections of Poisson's ratio $\nu(n)$ in the same planes. The color bars encode the magnitude of each property in GPa.

decreasing elastic moduli and Debye temperature in going from Mg_3AsBr_3 to Mg_3BiBr_3 suggests that the Mg_3BiBr_3 compound might exhibit even lower thermal conductivity, potentially beneficial for thermoelectric performance.⁷⁷

The elastic anisotropy of a material is crucial for understanding its mechanical behavior under complex stress states and has implications for various phenomena, including crack propagation, plastic deformation, and phase stability. Figures 8–10 show the anisotropies of Young's modulus, shear modulus, and Poisson's ratio in single crystals of the studied materials. For Mg_3AsBr_3 , the anisotropy ratios are 1.072 for Young's modulus, 1.089 for shear modulus, and 1.248 for

Poisson's ratio, while for Mg_3SbBr_3 , these values increase to 1.166, 1.205, and 1.576, respectively. For Mg_3BiBr_3 the anisotropy ratios are 1.182 for Young's modulus, 1.225 for shear modulus, and 1.639 for Poisson's ratio. The anisotropies of Young's and shear modulus are close to 1, and thus, Figure 8a,e, Figure 9a,e, and Figure 10 a,e show spherical 3D plots; also, the polar plots shown in panels b–d and f–h of Figures 8–10 further verify the symmetry. In the case of Poisson's ratio, the anisotropy value increases with pnictogen size, which can be seen from Figures 8i, 9i, and 10i. The polar projection of the symmetric directional dependence of Poisson's ratio can be seen for all three materials from Figure 8j–l, Figure 9j–l,

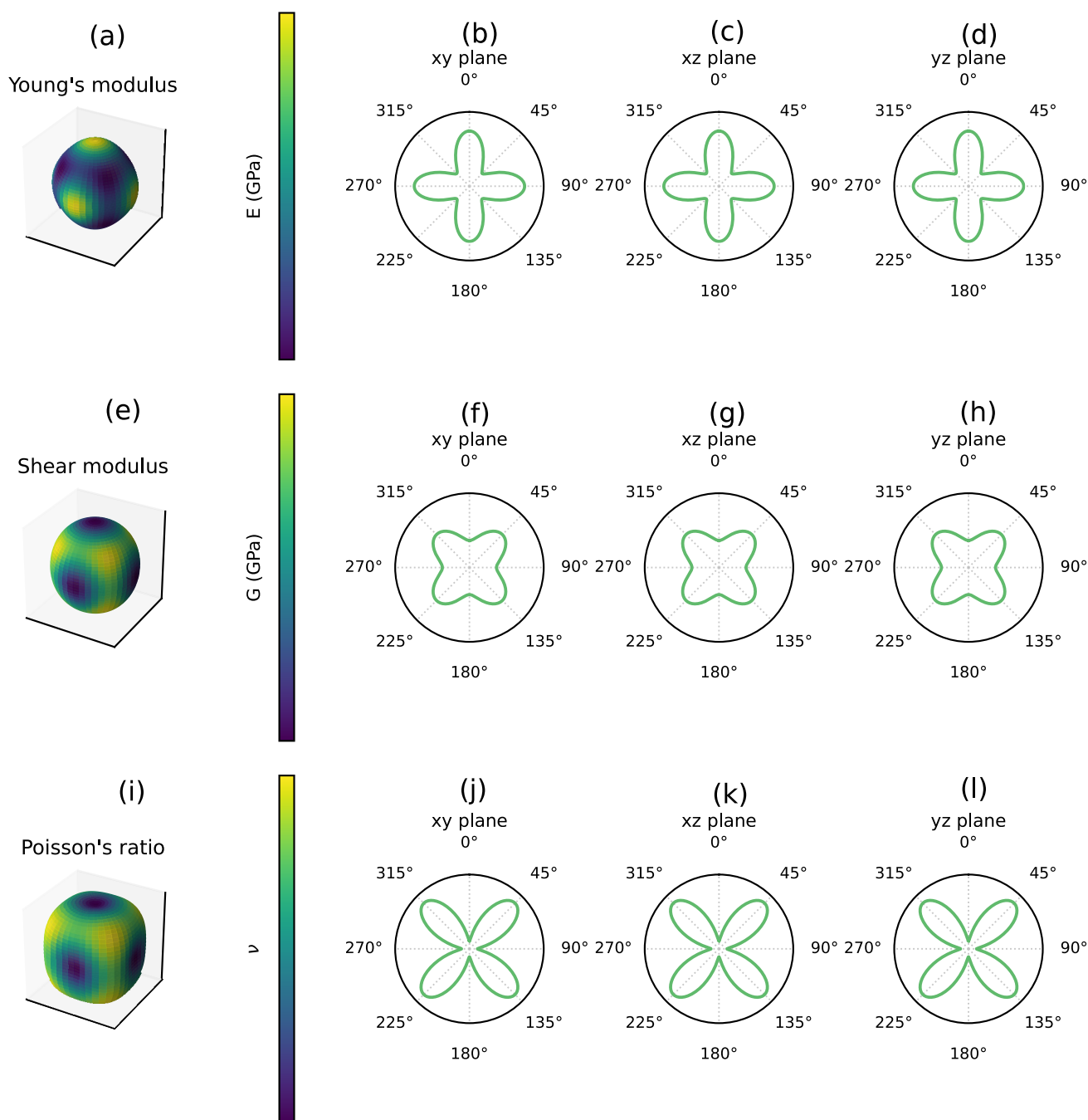


Figure 9. Orientation-dependent elastic response of cubic Mg_3SbBr_3 . (a–d) Three-dimensional surface and two-dimensional polar projections of the Young's modulus $E(n)$ in different crystallographic planes, respectively. (e–h) Corresponding representations of the shear modulus $G(n)$. (i–l) Polar projections of Poisson's ratio $\nu(n)$ in the same planes. The color bars encode the magnitude of each property in GPa.

and Figure 10j–l. This enhanced anisotropy with increasing pnictogen size in the compound indicates greater directional dependence of mechanical properties, which could influence the behavior under complex stress states and has implications for potential applications where directional mechanical response is important. Despite their cubic symmetry, all of the compounds exhibit some degree of elastic anisotropy, as quantified by the universal elastic anisotropy index (A^U) of 0.01 for Mg_3AsBr_3 , 0.04 for Mg_3SbBr_3 , and 0.050 for Mg_3BiBr_3 . These values are relatively low compared to those of many other cubic materials, indicating near-isotropic elastic behavior, which is advantageous for applications requiring uniform

mechanical response.⁷⁸ A more traditional measure of elastic anisotropy in cubic crystals is the Zener anisotropy ratio ($A = 2C_{44}/(C_{11} - C_{12})$), which equals 0.92 for Mg_3AsBr_3 , 0.83 for Mg_3SbBr_3 , and 0.82 for Mg_3BiBr_3 . These values, being close to unity—especially for Mg_3AsBr_3 —further confirm the relatively isotropic elastic nature of these materials, although the slightly lower value for Mg_3BiBr_3 suggests increasing anisotropy with larger pnictogen atoms.⁷⁹ The elastic anisotropy has important implications for microscopic deformation mechanisms and potential failure modes. Materials with significant elastic anisotropy often exhibit preferential slip systems and anisotropic fracture behavior. The relatively isotropic nature

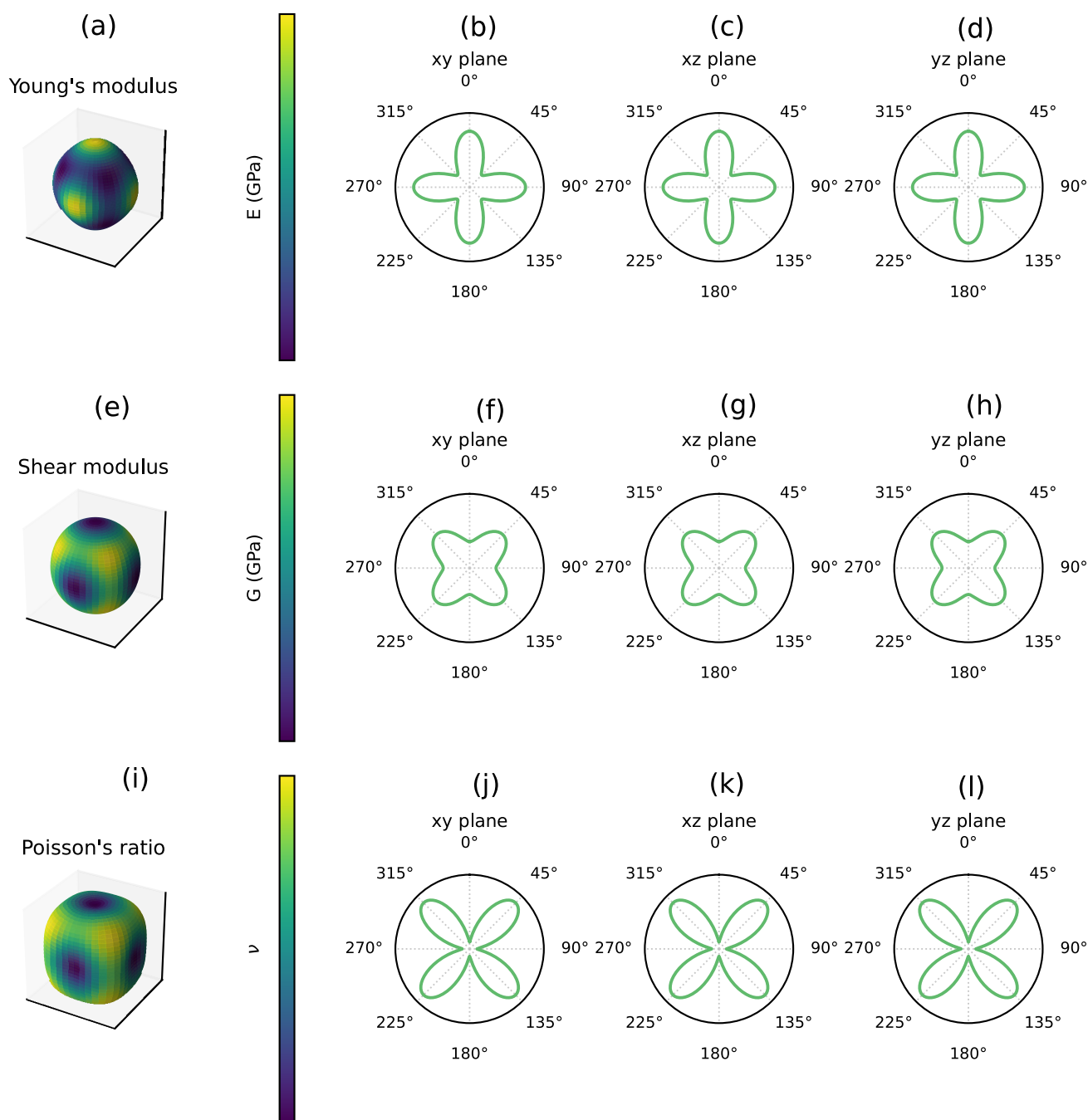


Figure 10. Orientation-dependent elastic response of cubic Mg_3BiBr_3 . (a–d) Three-dimensional surface and two-dimensional polar projections of the Young's modulus $E(n)$ in different crystallographic planes, respectively. (e–h) Corresponding representations of the shear modulus $G(n)$. (i–l) Polar projections of Poisson's ratio $\nu(n)$ in the same planes. The color bars encode the magnitude of each property in GPa.

of the Mg_3ZBr_3 compounds suggests more uniform mechanical response and potentially more predictable failure modes, advantageous for structural applications.⁸⁰

The decreasing elastic constants in Figure 11a, elastic moduli in Figure 11b, Vickers hardness in Figure 11c, and Zener parameter in Figure 11d can be observed. A high B/C_{44} ratio generally suggests better machinability, and for these materials, the B/C_{44} ratio increases with pnictogen atom size, as can be observed from Figure 11c. All these trends with increasing pnictogen size ($\text{As} \rightarrow \text{Sb} \rightarrow \text{Bi}$) can be attributed to the decrease of bond strength with increasing ionic radius in going from As^{3+} (0.58 Å) to Sb^{3+} (0.76 Å) to Bi^{3+} (1.03 Å).⁶⁵

The valence electron configurations (As , $4s^24p^3$; Sb , $5s^25p^3$; Bi , $6s^26p^3$) also influence bonding characteristics; the more diffuse orbitals of heavier pnictogens lead to weaker overlap with Br orbitals and consequently reduced bond strength and elastic moduli.⁸¹ The increasing polarizability in going from As to Sb to Bi results in more easily distorted electron distributions under stress, contributing to decreased elastic moduli and increased Poisson's ratio.⁷³

6. Device Characteristics of Mg_3ZBr_3 PIN Photodiodes

6.1. Structure and Material Parameters. Rectangular Mg_3ZBr_3 PIN devices, each 5 μm wide and normalized to an out-of-plane depth of 1 μm , were analyzed at 293 K using the

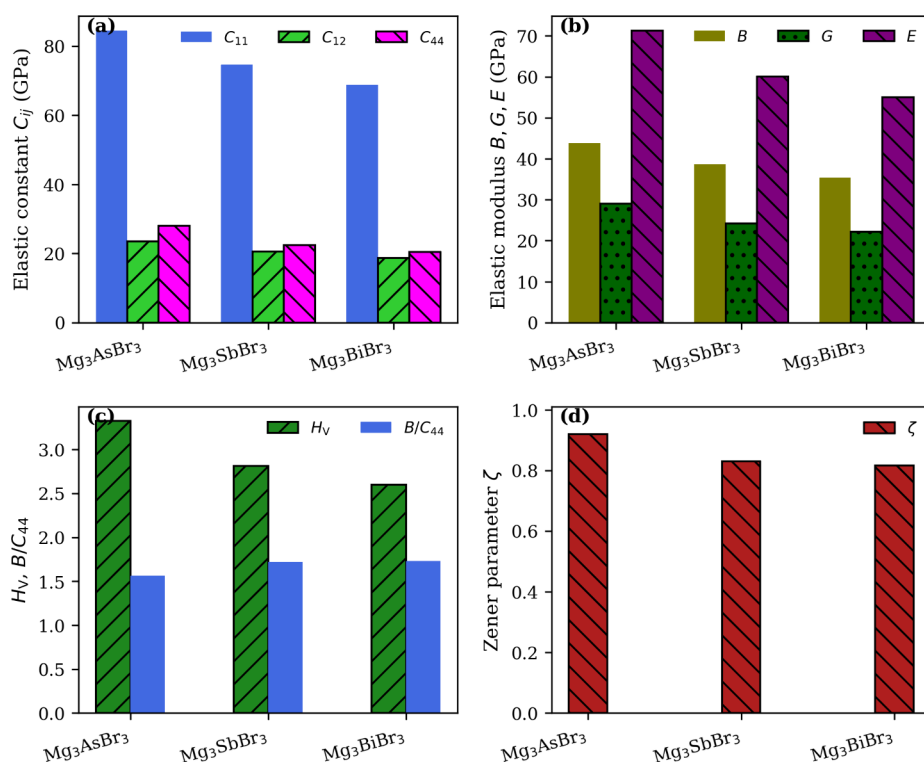


Figure 11. Calculated mechanical properties of Mg₃AsBr₃, Mg₃SbBr₃, and Mg₃BiBr₃: (a) single-crystal elastic constants C_{ij} (C_{11} , C_{12} , C_{44}); (b) bulk modulus (B), shear modulus (G), and Young's modulus (E); (c) Vickers hardness (H_V) and B/C_{44} ratio; (d) Zener anisotropy parameter (ζ).

Optoelectronics interface of COMSOL Multiphysics. The simulated stack—illustrated in Figure 12a,b—consists of p-

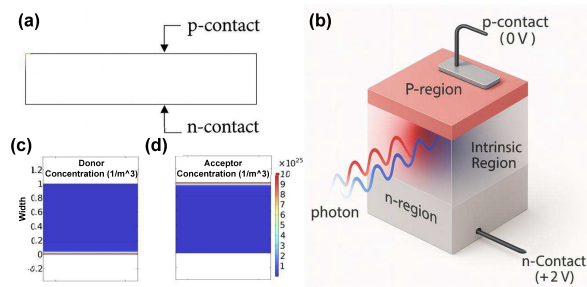


Figure 12. Schematic and simulated characteristics of the Mg₃ZBr₃ ($Z = \text{As, Sb, Bi}$) PIN photodiode under photoconductive mode. (a) The device geometry is a simple rectangle with a p contact on the top surface and an n contact on the bottom surface. (b) 3D structural schematic of the photodiode, highlighting the p, intrinsic, and n regions along with metal contacts and applied bias. (c) Donor concentration profile. (d) Acceptor concentration profile.

and n-type cladding layers separated by an intrinsic absorber. The p, i, and n regions are all based on the same Mg₃ZBr₃ composition (homojunction), so no conduction- or valence-band offsets are imposed at the internal p–i and i–n interfaces. A uniform dopant density of $1 \times 10^{20} \text{ cm}^{-3}$ was applied to the contact regions (Figure 12c,d).

Transport calculations employ the room-temperature mobilities listed in Table 5, a single midgap SRH trap with $\tau_n = \tau_p = 1 \text{ ns}$, and bulk band-to-band radiative and Auger recombination described by the coefficients B and $C_n = C_p$ (see below). No explicit surface-recombination velocities S_n or S_p were imposed at the contacts, so recombination is dominated

by these bulk processes. The electrostatics and transport follow the standard Poisson/continuity system:^{82,83}

$$\nabla \cdot (\epsilon \nabla \phi) = -q(p - n + N_D^+ - N_A^-) \quad (10)$$

$$\nabla \cdot \mathbf{J}_n = q(R - G), \quad \mathbf{J}_n = q\mu_n n \nabla \phi + qD_n \nabla n \quad (11)$$

$$-\nabla \cdot \mathbf{J}_p = q(R - G), \quad \mathbf{J}_p = q\mu_p p \nabla \phi - qD_p \nabla p \quad (12)$$

with Einstein relations $D_{n,p} = \mu_{n,p} k_B T/q$. Recombination includes SRH, radiative, and Auger terms:^{84,85}

$$R = R_{\text{SRH}} + R_{\text{rad}} + R_{\text{Auger}},$$

$$\begin{cases} R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \\ R_{\text{rad}} = B(np - n_i^2), \\ R_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2) \end{cases} \quad (13)$$

For the SRH channel we use a single midgap trap with $\tau_n = \tau_p = 1 \text{ ns}$. For halide-perovskite-like radiative recombination we adopt $B = 1.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$, within the experimentally established room-temperature range $k_2 \approx (0.6\text{--}14) \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$.^{86,87} Auger coefficients were set to $C_n = C_p = 1.0 \times 10^{-28} \text{ cm}^6 \text{ s}^{-1}$, consistent with reported $k_3 \approx (0.2\text{--}1.6) \times 10^{-28} \text{ cm}^6 \text{ s}^{-1}$ and first-principles/experimental determinations near $7.3 \times 10^{-29} \text{ cm}^6 \text{ s}^{-1}$.^{86,88} Reverse-bias LED operation was evaluated at $V_{\text{app}} = 2 \text{ V}$. A mapped 1D mesh (minimum element size $\lesssim \min\{L_D, L_T\}/10$) was used; a zero-bias equilibrium solve initialized the carriers, followed by stationary Newton iterations at the target bias with relative tolerance 10^{-6} . The dark current is the current that flows through the photodetector in the absence of incident light, primarily due to

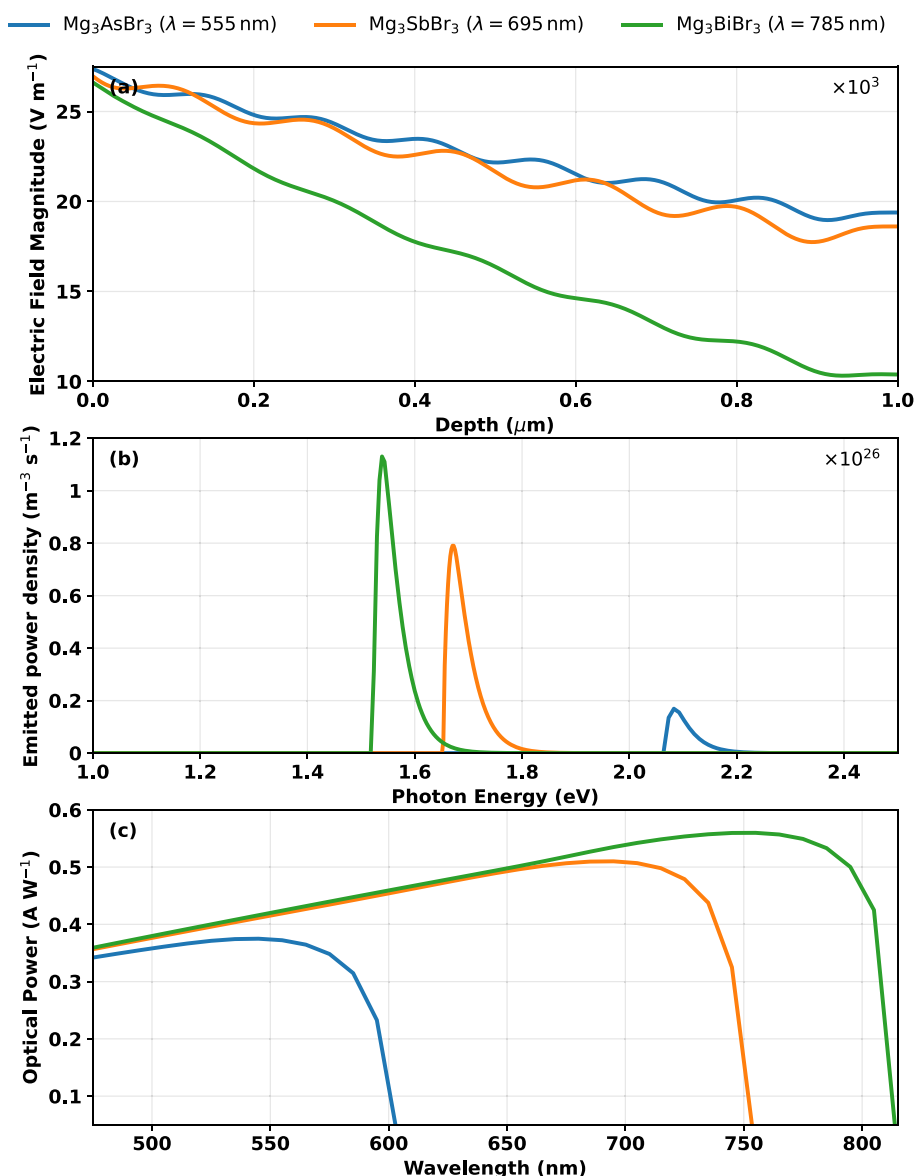


Figure 13. Simulated device characteristics of Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 PIN photodiodes. (a) Electric-field magnitude in the intrinsic layer at the wavelength of peak responsivity, showing an approximately linear decay consistent with a uniformly depleted $1 \mu\text{m}$ absorber under modest external bias. (b) Emitted power density as a function of photon energy. (c) Spectral responsivity $R(\lambda)$ under monochromatic illumination.

the thermal generation of carriers. In the present analysis we focus on the illuminated response and do not explicitly include dark-current noise contributions.⁸⁹

6.2. Spontaneous Emission and Recombination. The effectiveness of the PIN structure in photodiode devices stems from the sloping conduction- and valence-band profiles, which can be observed in Figure 13a–c, with the highest energy at the p contact and the lowest energy at the n contact. From the same figure, it can be seen that in the intrinsic region of the device the electron quasi-Fermi level lies below the conduction band and the hole quasi-Fermi level lies above the valence band. This energy alignment ensures that the conduction band is essentially empty and the valence band is filled, allowing efficient absorption of incoming photons. When light is absorbed and generates an electron–hole pair, the built-in electric field drives the electron toward the n contact and the hole toward the p contact.

Figure 13a presents the variation in the electric field magnitude across the device for different incident wavelengths. The strength of the field decreases during propagation, showing a nearly linear trend because of the $1 \mu\text{m}$ absorbing thickness; for a thicker layer, the characteristic exponential decay of the optical field would be more apparent.

Figure 13b shows the simulated spontaneous emission spectra from the Mg_3ZBr_3 PIN diodes under above-bandgap illumination (the same conditions used for the responsivity curves). Under reverse bias, a fraction of the photogenerated electron–hole pairs recombine radiatively in the intrinsic region, giving rise to photoluminescence (electroluminescence would occur under forward bias, but here the photon output is a result of optical pumping). The emission spectra confirm that each material's photon-emission onset corresponds closely to its bandgap energy. For the Bi-based diode, no emitted light is seen for photon energies below $\sim 1.52 \text{ eV}$, consistent with the transparency of the material below the band edge. Once the

incident photon energy exceeds E_g , radiative recombination can occur; Mg_3BiBr_3 shows an emission band peaking at around 1.55–1.65 eV. Analogous behavior is observed for the Sb and As devices. Mg_3SbBr_3 (gap ~ 1.65 eV) emits photons only above ~ 1.7 eV, with a peak around 1.7–1.8 eV. Mg_3AsBr_3 , having the largest gap, begins to emit at about 2.0 eV and peaks near 2.15 eV. In essence, each device emits a spectrum characteristic of its bandgap when pumped with superbandgap light, and the peaks of spontaneous emission align with the expected band-edge radiative transitions of these materials.

In the drift–diffusion framework used here, radiative recombination enters via the effective band-to-band term $R_{\text{rad}} = B(np - n_i^2)$ in the total recombination rate (see section 6.1). We do not explicitly resolve phonon-assisted indirect transitions at the microscopic level. Instead, the moderately indirect bandgaps obtained from the HSE06 calculations (section 4) are reflected in the choice of E_g and in the wavelength-dependent absorption coefficient used to construct the optical generation rate $G(x, \lambda)$. In this sense, the indirect nature of the transitions is treated phenomenologically through the effective coefficient B and the calculated absorption spectrum rather than by introducing explicit phonon-assisted recombination terms into the drift–diffusion model.

The simulations substantiate that Mg_3ZBr_3 family of halide perovskites furnish a tunable platform for near-IR to visible photodetection. By a judicious compositional choice, one can engineer both the cut-on wavelength and the peak responsivity without altering the device geometry. These quantitative insights furnish a roadmap for fabricating high-gain, low-dark-current Mg-based halide PIN photodiodes, and they lay the groundwork for subsequent experimental validation of this emerging semiconductor cohort.

6.3. Spectral Responsivity and Photocurrent. Figure 13c compares the simulated spectral responsivity $R(\lambda)$ —defined as the photocurrent output per incident optical power (A/W)—of the Mg_3AsBr_3 , Mg_3SbBr_3 , and Mg_3BiBr_3 PIN photodiodes under steady monochromatic illumination. The incident light power is fixed (per unit area), and the wavelength is swept from 875 to 475 nm. As expected, the Mg_3AsBr_3 device shows a sharp long-wavelength cutoff just below 840 nm, corresponding to its ~ 1.48 eV bandgap; virtually no photocurrent is generated for $\lambda > E_g/hc$. In contrast, the Sb- and Bi-based diodes remain photoresponsive at 875 nm (1.417 eV), since this photon energy exceeds their smaller bandgaps. For Mg_3BiBr_3 , with the narrowest gap, the responsivity is in fact highest at the longest wavelength in the range (near 875 nm), and $R(\lambda)$ gradually decreases toward the blue end of the sweep. This behavior reflects the trade-off between photon absorption probability and photon flux: at longer wavelengths (just above the band edge), the material absorbs photons less strongly but the fixed incident power corresponds to a higher photon flux. At shorter wavelengths, absorption in the $1\ \mu\text{m}$ i layer becomes nearly complete, but fewer incident photons (due to higher energy per photon) limit the photocurrent. It can also be observed that the peak shifts to longer wavelengths as E_g decreases in going from As to Bi. Specifically, Mg_3AsBr_3 reaches a peak at $R \approx 0.09$ A/W around $\lambda \sim 720$ –730 nm. Mg_3SbBr_3 shows a broader responsivity plateau peaking near ~ 800 nm with a slightly higher maximum R (on the order of 0.1 A/W). Mg_3BiBr_3 , which can absorb effectively throughout the entire 875–475 nm range, exhibits a relatively flat or gently sloping response

that tends to peak toward the red end; for instance, $R \approx 0.08$ –0.09 A/W at 875 nm and gradually diminishes to ~ 0.05 –0.06 A/W by 500 nm. We note these values are lower than those of optimized or thicker photodiodes; e.g., single-crystal or tandem perovskite photodetectors often report $R \sim 0.3$ –0.6 A/W in the visible range.⁸⁹ The modest R here is attributable to the simple device design (only $1\ \mu\text{m}$ absorption path, no reflective or optical concentrator layers) and the assumption of unity carrier collection efficiency without internal gain. Nonetheless, the extension of spectral sensitivity into the infrared for Sb- and especially Bi-based Mg_3ZBr_3 is a clear advantage.

CONCLUSIONS

First-principles calculations identify cubic perovskite Mg_3ZBr_3 ($Z = \text{As, Sb, Bi}$; $Pm\bar{3}m$) as a chemically simple, lead-free halide family with predictable trends across structure, lattice dynamics, and optoelectronic response. After relaxation, the lattice parameter and equilibrium volume increase in going from As to Sb to Bi, consistent with the larger pnictogen size. Phonon dispersions contain no imaginary branches at 0 K for Mg_3AsBr_3 and Mg_3SbBr_3 , consistent with a dynamically stable cubic phase. Hybrid functional band structures yield indirect gaps between ~ 2.06 and ~ 1.52 eV across the pnictogen series, with lighter electrons than holes along the fundamental transition, suggesting electron-favored transport. Frequency-dependent optical functions follow the electronic trends. The static dielectric constant rises from ~ 4.8 to ~ 5.9 , and the near-IR refractive index lies near 2.2 to 2.5 for these compounds.

Elastic tensors satisfy the Born criteria for all three compositions. Hill-averaged moduli decrease smoothly down the series (bulk modulus $\approx 44 \rightarrow 35$ GPa; Young's modulus $\approx 71 \rightarrow 55$ GPa), accompanied by lower Debye temperatures. Quasi-harmonic mode Grüneisen parameters are ($\bar{\gamma} \gtrsim 2.7$ –3.3), consistent with strong anharmonicity and low predicted lattice thermal conductivity.

Device-level drift–diffusion simulations of p–i–n stacks reproduce band-edge-limited spectral cut-ons and rank-order the responsivity with composition, confirming that this series can cover near-IR to visible wavelengths without changing the device geometry. These results point to applications in photodiodes and radiation detection, and they motivate measurement of thermal transport and finite-temperature phase behavior to test the predicted anharmonicity and stability margins.

ASSOCIATED CONTENT

Data Availability Statement

The data underlying the findings of this study are openly available in the Zenodo repository at [10.5281/zenodo.17982237](https://doi.org/10.5281/zenodo.17982237).

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c11440>.

Figures S1–S5 and Table S1 (Bader charges) (PDF)

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[§]M.M. and M.M.H. contributed equally. M.M. designed the study, performed the calculations, analyzed the data, and wrote the manuscript. M.M.H. and A.K. administered the project and supervised the investigations by guiding the computations and analyzing the data. All of the authors reviewed the manuscript.

Notes

The authors declare no competing financial interest.

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